Chapter 1

Introduction

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This Scientific and Technical Report (STR) is the summary of the work of the IWA Task Group on Benchmarking of Control Strategies for Wastewater Treatment Plants. As will be explained in Chapter 2, this Task Group has a long history. However, before describing this history and the results of the Task Group in more detail, we would first like to use this introduction to provide more insight into a number of basic issues related to the family of Benchmark Simulation Models (BSMs), which are the main 'products' of this Task Group. In order to do this, we will seek answers to a few basic questions: What is meant by a 'Benchmark Simulation Model'? What is the purpose of the BSMs? Who should use the BSMs? How should a BSM be used? Who has been involved in the development of the BSMs? And last but not least, how should this STR be read?

1.1 WHAT IS MEANT BY A 'BENCHMARK SIMULATION MODEL'?

When checking a dictionary, a benchmark is defined as a measure of reference to be used in a test. In computer science a benchmark is a reference performance to which the relative performance of hardware or software can be compared. In process modelling and control, a benchmark is defined as a plant model and associated control strategy that can be used as a reference point for simulation-based comparison of control strategies (Downs & Vogel, 1993). Such a simulation benchmark is not associated with a particular simulation platform. Direct coding (e.g., C/C++, FORTRAN) as well as commercial simulation software packages can be used. In this case, the purpose of the simulation protocol was to create a tool that could guarantee that different users obtain exactly the same results when running the simulation model. The main 'products' of this Task Group are WWTP simulation models (Chapter 4), a simulation protocol for these simulation models (Chapter 7) and a set of benchmarking evaluation criteria for objective control evaluation (Chapter 6). All these items together form the benchmark simulation model platform. However, it should be emphasised as well that a major result of the Task Group work is a set of verified unit process models and tools that are applicable to WWTP simulation studies in general. The Activated Sludge Model No. 1 (ASM1), Takács secondary clarifier model and Anaerobic Digestion Model No. 1 (ADM1) among others were all verified before including those unit process models in the BSM platform. More details on these models will be provided in Chapter 4. Other tools presented in Chapter 4, such as a dynamic influent

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pollution disturbance scenario generator (DIPDSG) model, AS-AD interface models, sensor and actuator (fault) models and a risk assessment module for microbiology related settling problems have also resulted from the work of the Task Group.

1.2 WHAT IS THE PURPOSE OF THE BENCHMARK SIMULATION MODELS?

The activated sludge process aims to achieve, at minimum cost, sufficiently low concentrations of biodegradable matter and nutrients in the effluent together with minimal sludge production. To do this, the process has to be controlled. Many control strategies have been proposed in the literature; however, the literature does not provide a clear basis for comparison of these strategies because of the many confounding influences that have an impact on the system. Many of these influences are easily recognised. For instance, physical characteristics of the process can have an impact on process performance and this makes the comparison of strategies applied to different reactor layouts difficult. As well, the influence of a control strategy on process performance is expected to vary with different disturbances, thus the disturbances used to test the control strategy become important. Also complicating the evaluation is the lack of standard evaluation criteria. Effluent requirements are for example often location specific, which makes it difficult to judge the particular influence of an applied control strategy from a reported performance increase. Controversies that result from the resulting subjective reports reinforce the need to devise an effective and unbiased evaluation method that can be used to judge the utility of different control strategies.

From a practical standpoint, it is not reasonable to experimentally test and verify the effectiveness of all reported control strategies and often the assessment of these control strategies is confounded by the multi-faceted nature of the process under study. Alternatively, given a standardised procedure, it is possible to efficiently evaluate numerous strategies through realistic/dynamic computer simulations. Simulations provide a cost-effective means for the evaluation of control strategies, but the unlimited number of simulation permutations makes the need for a standardised protocol very important if different strategies (and different simulation results) are to be compared. Each control strategy must be simulated under the same conditions to ensure unbiased comparisons. Validation of the computer simulations is difficult without supporting experimental or full-scale data, but the value of the work is enhanced through the use of accepted activated sludge models. Because appropriate simulation tools for the activated sludge process are available this approach has numerous advantages, but still there is a need for a standardised procedure. To this end, there has been an effort to develop a standardised simulation protocol – a 'simulation benchmark'.

The BSM platform was originally intended exclusively for the simulation-based comparison of WWTP control strategies. This was later extended to include the comparison of WWTP monitoring strategies as well (Chapters 4, 5, 6 and 7). Process monitoring is the activity by which collected data are analysed to find process deviations, failures and faults. Process monitoring also involves isolation of variables contributing to the deviations, facilitating further analysis of the problems. Again, many monitoring methods have been proposed in the literature but an unbiased evaluation of their performance in a WWTP was not available even though it was highly desired. Consequently, a standardised simulation protocol, verified models and evaluation criteria were established to address this deficiency.

1.3 WHO SHOULD USE THE BENCHMARK SIMULATION MODELS?

It is not that easy to describe a specific target group for the BSM platform. In fact, the models and tools described in this STR should appeal to a broad audience – that audience will be called 'benchmark users' in the rest of the STR – both from industry and academia with a general interest in WWTP modelling. This is best illustrated with a few examples. The benchmark simulation models can be used for objective

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simulation-based comparison of control strategies. As such, users from academia might find it interesting to develop a control strategy that yields the best overall performance. They may also use it to train their students on the basics of process control. A consultant, on the other hand, might use these models to demonstrate the use of control strategies to a potential customer. A benchmark user might also be interested in using only part of the benchmark simulation model (e.g., the evaluation criteria, or sensor and actuator models, or the influent generator). And of course, individual unit process models can be used in any simulation study as they all are now verified by the Task Group and are distributed for free when requested.

1.4 HOW SHOULD THE BENCHMARK SIMULATION MODELS BE USED?

The BSM platform is a standardised simulation protocol. If it is the intention of the benchmark user to work with the BSM platform and to compare simulation results with other benchmark users, then the standardised simulation protocol should be followed. A space limitation in this STR means that it was not possible to provide a description of all the minute details of the BSMs. Rather, this publication provides a detailed overview of the BSMs and the various tools. For more specific details, the reader is referred to the detailed technical reports and BSM computer code which accompany the STR.

The benchmark user is of course free to modify the models and tools provided by the Task Group. However, such modifications should be mentioned clearly and documented properly when publishing results in order to allow a fair comparison of results.

1.5 WHO HAS BEEN INVOLVED IN THE DEVELOPMENT OF THE BENCHMARK SIMULATION MODELS?

As will be explained in Chapter 2 it has taken several years to develop the BSM platform and the Task Group 'products' are truly the result of a group effort. Over the years, many people have contributed to the BSM platform development. These people were not all part of the Task Group and many did not specifically contribute to this publication. However, those individuals should be fully acknowledged for their work and input. The Task Group has tried to do so in the preface to this publication, knowing that we will probably never be able to name all the people that have been involved in the development over the years. In the rest of this STR, we will collectively refer to the BSM developers as the 'benchmarkers'.

1.6 HOW SHOULD THIS SCIENTIFIC AND TECHNICAL REPORT BE READ?

This STR is only a summary of the work of the Task Group and can be used to get a rapid overview of the results. The detailed technical reports were written by the benchmarkers while developing, implementing and validating the specific models and tools, and are as such the most detailed information provided by the Task Group. To supplement that reading, the reader can consult the model code provided by the Task Group in order to get a deeper understanding of the more technical issues. Finally, a benchmark user can find inspiration in the literature, as a significant number of scientific papers published during the years have been devoted to the use of the BSMs. These papers are listed in Technical Report No. 15. As a courtesy to the original contributors, benchmark users are asked to refer to those literature sources when publishing their own work, in order to properly acknowledge the work done by the different benchmarkers.

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Chapter 2

Benchmark overview

M.-N. Pons, C. Rosen and U. Jeppsson

This chapter of the STR provides a concise historic overview of the key events and decisions that have resulted in the development of the different Task Group WWTP simulation benchmarks.

2.1 BENCHMARK SIMULATION MODEL NO. 1

In chemical engineering, the Tennessee-Eastman plant has been used extensively in research and teaching for process optimisation, process control and process monitoring. The plant is a process modelling and control benchmark describing a reactor/separator/recycle arrangement involving two main simultaneous gas-liquid exothermic reactions with associated operating objectives, process control objectives and process disturbances (Downs & Vogel, 1993).

Undoubtedly, the concepts behind the Tennessee-Eastman plant have been a major source of inspiration for the development of the WWTP simulation benchmarks described in this STR. In fact, the benchmarking idea was first mentioned by Professor Bengt Carlsson (Uppsala University, Sweden) at IWA's Instrumentation Control and Automation Conference in Hamilton (Canada) in 1993. This idea was inspired by the Control System Society in the USA related to flight control systems. The development of the first simulation benchmark (BSM1) was thus initiated two decades ago. The initial benchmarking concepts were largely defined in parallel with the work reported by Spanjers *et al.* (1998) and Copp *et al.* (2002), who developed and applied an activated sludge benchmark that focused on respirometry-based control. The concepts for this respirometry-based control benchmark were presented at a COST (European Cooperation in Science and Technology) 682 Working Group No. 2 meeting in Florence (Italy) in November 1996, and were very helpful as a starting point for the BSM1 development. A meeting in Wageningen (The Netherlands) in April 1997 was the true starting point for the BSM1 definition: the first model equations were written down and tasks to develop the main parts of BSM1 (performance evaluation, testing, influent definition, …) were assigned. At that meeting, the benchmark was defined as '*a protocol to obtain a measure of performance of control strategies for activated sludge plants based on numerical, realistic simulations of the controlled plant*'. At that time a number of strategic decisions were taken and most of those decisions have had a large impact on the final version of BSM1. It was decided that specific control strategies should not be included in the benchmark definition. Moreover, the benchmark should not

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simply be simulation code, but also should include definitions of influent disturbances and performance evaluation criteria. It was decided that the first layout would be aimed at carbon/nitrogen removal in a series of tanks. Discussion groups were set up to decide upon the final clarifier model, the influent description and the performance assessment procedure. The full set of equations, all the parameter values, the procedure to be applied to verify the code and simulation results were all made available for use and discussion on the dedicated website http://www.benchmarkwwtp.org/ organised within the COST Action 682. The benchmarkers exchanged results electronically with Dr. Ulf Jeppsson from Lund University (Sweden) who was in charge of coordinating the information.

A subsequent meeting was held in Ljubljana (Slovenia) in April 1998. The main issues discussed in Slovenia were the size of the clarifier, the initialisation procedure, the influent description and the performance assessment. Implementation of the benchmark was discussed in Lund (Sweden) in August 1998. At that time, the benchmark had been implemented on six platforms: three implementations were coded from scratch (FORTRAN, C and Matlab/Simulink) and three were implemented in commercial WWTP software packages (GPS-X, Simba and WEST). Ongoing ring-tests revealed large discrepancies in computation time. Furthermore, these ring-tests revealed errors in the implementation code (in all platforms) and potential problems related to algebraic loops (due to the recycle loops in the plant). This ring-testing exercise also identified the impact of subtle differences in the implementation of the models. That is, not all the models were input into the commercial simulators in exactly the same way (aeration, soluble components in the settler, . . .) and the ring-testing identified the impact of these differences. Implementing code from scratch allowed more flexibility as the written equations were translated precisely into code, but the disadvantage was that the implementation was more prone to errors.

The meeting in Sweden was the last one under the umbrella of COST Action 682 and so the benchmark development migrated to COST Action 624 'Optimal Management of Wastewater Systems'. A meeting in Grenoble (France) (March 1999) was devoted to refinements of the basic benchmark. All the details were made available on the website and were published in a 'benchmark manual' (Copp, 2002) documenting BSM1. When preparing the document, a substantial effort went into verifying the steady state and dynamic output data included in the description. The manual summarises the various tested implementations with helpful hints for new benchmark users. Results were verified using BioWin, EFOR, GPS-X, Matlab/Simulink, Simba, STOAT, WEST and a user-defined FORTRAN code (Alex *et al.* 1999; Pons *et al.* 1999; Copp, 2002).

When COST Action 624 came to an end in 2004, the BSM1 development continued as the IWA Task Group on Benchmarking of Control Strategies for Wastewater Treatment Plants. At this point, the main focus was the definition of an updated set of evaluation criteria and a set of standardised sensor models. Some adjustments to BSM1 were also made to improve its compatibility with BSM1_LT and BSM2 (Sections 2.2 and 2.3, respectively), both of which were subsequently developed by this Task Group.

During the entire BSM1 development period, a number of MSc and PhD studies benefited from its existence and numerous scientific papers have been published in which the simulation protocol or the tools that had been produced as a result are a central ingredient (Technical Report No. 15).

2.2 BENCHMARK SIMULATION MODEL NO. 1 LONG-TERM

Although a very flexible tool, BSM1 is not intended for long-term evaluations. In the BSM1 definition, the temperature is constant and the evaluation period is limited to one week for three different weather scenarios (Copp, 2002). However, it was realised that many of the control actions at a WWTP have an effect on the process over longer time scales: sludge age control, equalisation basin control and sludge storage to mention a few. Also, BSM1 is mainly based on ideal sensors and actuators, because short-term

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evaluations do not allow for realistic equipment (sensor/actuator) modelling, including failures, drift and maintenance, as these typically appear over longer time scales than one week.

The Benchmark Simulation Model No. 1 Long-Term (BSM1_LT) addresses these BSM1 shortcomings and BSM1_LT also allows for process monitoring. Process monitoring is closely linked to process control. Process monitoring in this context means tracking measurement variables to detect process deviations, failures and faults. Process monitoring also involves the isolation of variables contributing to the deviations, facilitating further analysis of the problems. Monitoring of wastewater treatment operations has in the last decade become an intensive area of research and many different methods have been proposed (Rosen *et al.* 2003). Unfortunately, before the development of BSM1_LT there was no objective way to compare the success of methods for wastewater treatment monitoring, as researchers generally have used real data specific for a certain plant and, thus, not generally available to others. The BSM1_LT is intended to resolve this problem and is the first simulation platform that allows the objective comparison of WWTP monitoring methods.

The basic BSM1_LT concepts were discussed in detail during a COST Action 624 meeting in Lund (Sweden) in December 2003, and were then presented at the AutMoNet 2004 conference in Vienna (Austria). Those basic BSM1_LT concepts were published in Rosen *et al.* (2004), and have been developed further under the umbrella of this Task Group. First, a long-term influent file consistent with the previously developed BSM2-influent was created (Chapter 4). Existing BSM1 models were extended to include temperature dependency (see also Chapter 4). Sensor models that adopted the principles and models described by Rieger *et al.* (2003) were implemented. Models for sensor and actuator faults (Rosen *et al.* 2008) were also developed and incorporated. BSM1_LT required novel and suitable performance evaluation criteria (Corominas *et al.* 2011), and this proved to be one of the most difficult tasks to accomplish during the development of BSM1_LT. Ring-tested implementations of the BSM1_LT are available in Matlab/Simulink and WEST.

2.3 BENCHMARK SIMULATION MODEL NO. 2

A further limitation of BSM1 is that it does not allow for evaluation of control strategies on a plant-wide basis. BSM1 only includes an activated sludge system and a secondary clarifier. Consequently, only local control strategies can be evaluated. During the last decade the importance of integrated and plant-wide control has been emphasised by the research community and the wastewater industry is starting to realise the benefits of such an approach. A WWTP should be considered a unit, where primary/secondary clarification units, activated sludge reactors, anaerobic digesters, thickeners, dewatering systems and so on are linked together and need to be operated and controlled not only on a local level as individual processes but also by supervisory systems taking into account all the interactions between the processes. Otherwise, sub-optimal performance will be an unavoidable outcome leading to reduced effluent quality and/or higher operational costs.

It was the intent of the Benchmark Simulation Model No. 2 (BSM2) to take these issues into account. BSM1 was updated by adding wastewater pre-treatment and a sludge train including anaerobic digestion (Chapter 3). To allow for a more thorough and extended evaluation, the benchmark evaluation period was extended to one year (compared to one week in BSM1). The slow dynamics of anaerobic digestion processes also necessitated a prolonged evaluation period. With this extended evaluation period, it was reasonable to include seasonal effects on the WWTP in terms of temperature variations. The influent data files included with BSM1 (dry, storm and rain weather data) have been used extensively by researchers. However, for BSM2 the extended dynamic influent is generated with a phenomenological influent model (Gernaey *et al.* 2011).

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The BSM2 concepts were first discussed in detail during a COST Action 624 meeting in Lund (Sweden) in December 2003 and were then presented at WaterMatex2004 in Beijing (China) and subsequently published by Jeppsson *et al.* (2006). Following the Lund meeting in 2003, the BSM2 development was coordinated by Dr. Ulf Jeppsson (Lund University, Sweden). Development continued under the umbrella of the Task Group starting in May 2005. The first BSM2 layout was based on the ideas outlined in Jeppsson *et al.* (2006). It was implemented and verified in accordance with the plan and presented at WaterMatex2007 in Washington D.C. (USA) (Jeppsson *et al.* 2007). Major milestones in the BSM2 development were the implementation and verification of the Anaerobic Digestion Model No. 1 (ADM1, Batstone *et al.* 2002; Rosen *et al.* 2006), the development of phenomenological influent models (Gernaey *et al.* 2011) and the development of ASM1 → ADM1 and ADM1 → ASM1 model interfaces (Nopens *et al.* 2009).

The BSM2 layout presented at WaterMatex2007 included 15 simple demonstration cases, both with and without active controllers, and was aimed at investigating how the evaluation criteria captured various operational conditions. It was however revealed that: (1) the evaluation criteria were not very sensitive to the different cases tested; and, (2) the very highly loaded system, which was deliberately adopted, limited what could be accomplished by active control. Indeed, it was shown that active control has its limitations and will not be able to significantly improve the performance of a highly overloaded plant. Interestingly, the high nitrogen load which was causing some of the issues was associated with the reject water, which was not present or accounted for in the BSM1 case. The final BSM2 layout was presented by Nopens *et al.* (2010). This layout (compared to the earlier versions) included: (1) a reduced N load to compensate for the contribution of the reject water; (2) increased activated sludge tank volumes, in order to obtain a low/medium loaded WWTP that can benefit from process control; and, (3) modifications of the evaluation criteria. The new criteria made a distinction between nitrate and ammonium nitrogen and reduced the dominating effect of aeration that had been observed in BSM1 and earlier versions of the BSM2. Ring-tested implementations of the BSM2 are available in Fortran, Matlab/Simulink, Simba and WEST.

2.4 THE BENCHMARK SIMULATION MODEL SET

From the outset, the BSM1, BSM1_LT and BSM2 protocols have been developed to help with the unbiased evaluation of control-related simulations. Through the years, issues have been identified and short-comings in the models have been debated. The current protocols represent the state-of-the-art and encompass a scope meant to be representative of most wastewater treatment control applications. However, with each new application, and there have been hundreds, comes suggestions for improvements and although this STR deals only with these three protocols, further developments are fully expected and encouraged in the future.

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Chapter 3

Benchmark plant description

U. Jeppsson, J. B. Copp, K. V. Gernaey, M.-N. Pons and P. A. Vanrolleghem

This chapter of the STR provides a brief overview description of the different WWTP simulation benchmarks that are distributed by the Task Group. In practice, this chapter provides a description of the benchmark-specified equipment in each benchmark plant. Furthermore, two reference configurations have been defined for each benchmark plant. These reference configurations are referred to as the 'open-loop' and 'closed-loop' configurations. The 'open-loop' configuration (without any control) provides a reference point to compare the performance of a control and/or monitoring strategy and the 'closed-loop' configuration is the same configuration with a defined and active default control strategy added to it. This chapter describes the open-loop reference configuration. A full description of the models, evaluation criteria and simulation procedures are described in subsequent chapters. The closed-loop configurations (i.e., the default control strategies) are described in Chapter 5.

3.1 BENCHMARK SIMULATION MODEL NO. 1

The purpose of the BSM1 system was to define a tool that would allow for the objective and unbiased comparison of carbon and nitrogen removal activated sludge control strategies. The BSM1 plant is a simple layout consisting of an activated sludge tank and a secondary clarifier as schematically shown in Figure 3.1. For the open-loop configuration, both liquid flow rates and aeration intensity are fixed.

The physical attributes of the open-loop BSM1 layout (Figure 3.1) are defined as follows:

- The temperature of the influent and the mixed liquor is constant at 15°C
- The average dry weather influent flow rate to the plant is 18 446 $m^3 \cdot d^{-1}$
  - step feed of the influent is possible, but the open-loop configuration has all of the influent flow directed into the first unaerated tank
  - influent composition is described in Chapter 4
- The activated sludge tank consists of 5 tanks-in-series, with a total biological volume of 5 999 m^3
  - tanks 1 and 2 are 1 000 m^3 each, tanks 3, 4 and 5 are 1 333 m^3 each
  - tanks 1 and 2 are unaerated in open-loop, but fully mixed

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  - tanks 3, 4 and 5 are aerated
  - for the open-loop case, the oxygen transfer coefficients (K_La) are fixed
  - for tanks 3 and 4 the coefficient is set to 240 d^{-1} (10 h^{-1})
  - for tank 5 the coefficient is set to 84 d^{-1} (3.5 h^{-1})
  - the DO saturation concentration in tanks 3, 4 and 5 is set to 8 g $O_2 \cdot m^{-3}$
- The secondary clarifier has a volume of 6 000 m^3 (area of 1 500 m^2 and a depth of 4 m)
  - the feed point to the settler is at 2.2 m from the bottom.
- Waste activated sludge (WAS) is pumped continuously from the secondary settler underflow at a default flow rate of 385 $m^3 \cdot d^{-1}$
- There are two recycle flows
  - a 55 338 $m^3 \cdot d^{-1}$ internal recycle from the 5th to the 1st tank
  - a 18 446 $m^3 \cdot d^{-1}$ RAS recycle from the secondary settler underflow to the 1st tank.

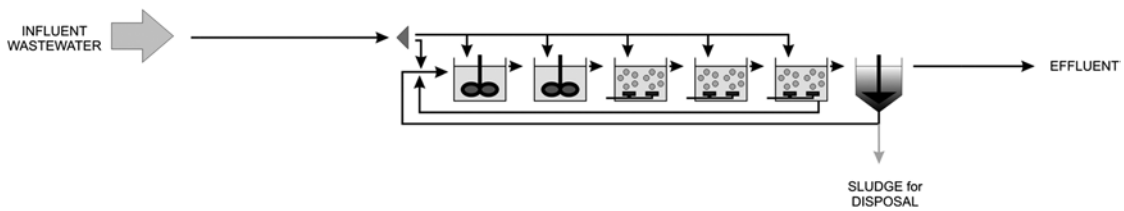

Figure 3.1 Schematic representation of the *BSM1* plant design.

3.2 BENCHMARK SIMULATION MODEL NO. 1 LONG-TERM

The basic features of the BSM1_LT plant layout are identical to BSM1. The differences between BSM1_LT and BSM1 are:

- The evaluation period is extended to 52 weeks (1 year)
- Long-term dynamics are introduced
  - the extended influent includes dry weather as well as rain events, [i.e., the BSM1 approach of using separate 14-day influent files (one for dry weather, one for rain, and one for a storm event) was altered so that one influent file included all of these types of events over an extended period]
  - the average influent flow rate for the evaluation period is 20 504 $m^3 \cdot d^{-1}$
  - the influent and mixed liquor temperatures vary with time
  - details on the influent dynamics and composition are provided in Chapter 4
- Sensor, actuator and process faults are included and described in Chapter 4.

3.3 BENCHMARK SIMULATION MODEL NO. 2

The main driver for the development of BSM2 was to extend the BSM concept to a long-term and plant-wide context. Following the development and subsequent experience with BSM1, it was realised that the lack of sludge handling in BSM1 could limit and bias the comparison of different control strategies, even though things like sludge production were being considered in the evaluation criteria. The BSM2 concept was developed to address these perceived short-comings. BSM2 (Figure 3.2) is based on BSM1 for the biological treatment, but, BSM2 includes a primary clarifier, a thickener for the sludge wasted

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from the BSM1 secondary clarifier, an anaerobic digester for treatment of thickened secondary sludge and solids wasted from the primary clarifier and a dewatering unit for the digested sludge. The effluent liquid streams from the thickening and dewatering steps are recycled back to the inlet of the primary clarifier. With these additions, different control handles such as pumps and valves have been included (Figure 3.2).

The details of the open-loop plant are as follows:

- The influent and mixed liquor temperatures vary with time
- The average influent flow rate for the evaluation period is 20 648 $m^3 \cdot d^{-1}$
  - similar to BSM1_LT, this influent includes dry weather as well as rain and storm events and covers an extended period of time (1 year)
  - details of the influent dynamics and composition are provided in Chapter 4
- The primary clarifier has a volume of 900 m^3 (area of 300 m^2, depth of 3 m)
  - the default underflow from the primary clarifier is 147.6 $m^3 \cdot d^{-1}$

> NOTE: While developing BSM2, it was discovered that the BSM1 biological tanks were severely overloaded with nitrogen partly as a consequence of the nitrogen rich reject water recycled from the dewatering step. As a result, for BSM2, the influent nitrogen load was reduced and tank volumes were re-evaluated on the basis of the design guidelines from both the German Association for Water Economy, Wastewater and Waste (ATV A131, 2000) and the US Environmental Protection Agency (Harris *et al.* 1982), as explained in Nopens *et al.* (2010). Both guidelines suggested that the volume of the activated sludge tanks be increased considerably.

- The activated sludge tank is configured as five (5) tanks-in-series with a total volume of 12 000 m^3
  - the first two unaerated tanks have a volume of 1 500 m^3 each
  - the three aerobic tanks have a volume of 3 000 m^3 each
  - the default K_La coefficients in the aerobic tanks are fixed, 120 d^{-1} (5 h^{-1}) for tanks 3 and 4 and 60 d^{-1} (2.5 h^{-1}) for tank 5
- Two activated sludge recycles are included
  - a 20 648 $m^3 \cdot d^{-1}$ RAS recycle from the secondary settler underflow to the 1st tank
  - an internal recycle with a default flow rate of 61 944 $m^3 \cdot d^{-1}$ from the 5th to the 1st tank
- Supplemental carbon is dosed to the first anoxic tank, at a default flow rate of 2 $m^3 \cdot d^{-1}$ and concentration of 400 kg COD $\cdot$ m^{-3}
- The secondary clarifier has a volume of 6 000 m^3 (area of 1 500 m^2, depth of 4 m)
  - the feed point to the clarifier is at 2.2 m from the bottom
- The anaerobic digester (AD) has a liquid volume of 3 400 m^3 and gas volume (headspace) of 300 m^3
- The default WAS flow rate is 300 $m^3 \cdot d^{-1}$ and is fed into the thickener
- The thickener and the dewatering units have no volume
  - these units are considered to be ideal point separators that, at steady state, produce concentrated sludge flows of 7% and 28% TSS, respectively
  - the underflow from the thickener is 30.9 $m^3 \cdot d^{-1}$
  - the dewatering generates 9.6 $m^3 \cdot d^{-1}$ of concentrated sludge and 168.9 $m^3 \cdot d^{-1}$ of reject water
- The reject water storage tank has a volume of 160 m^3
  - in the open-loop case the storage tank is not used, that is, the reject water bypasses the storage tank and is combined with the influent before the inlet of the primary clarifier.

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Chapter 4

Benchmark models

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The benchmark simulation models (BSMs) are the result of the combination of a number of sub-models that have been used to mathematically describe each process in the different benchmark layouts (Chapter 3). This chapter provides a brief description of those benchmark sub-models. Some of these models are specific to particular unit processes (i.e., ASM1) and others are globally applicable (i.e., temperature).

The first sections of this chapter address a new influent wastewater generation model specifically developed for the BSMs, but universally applicable. This is followed by sections on individual unit process models, sensors and actuators, and inhibition and toxicity modelling. The final sections address more general concepts including a new approach aimed at predicting the possibility of settling problems and finally the methods used for modelling temperature and the effect of temperature on kinetic parameters.

Table 4.1 shows the various unit process models, and the benchmark that they are associated with.

Table 4.1 Modelling capability included in the three BSMs. A '✓' signifies that the benchmark includes that behaviour or capability, but its implementation is simulation-specific.

	Benchmark model		
	BSM1	**BSM1_LT**	**BSM2**
Influent	14-day data files	model generated 609-day data file	model generated 609-day data file
Activated sludge	ASM1	ASM1	ASM1
Anaerobic digester	–	–	ADM1
Primary clarifier	–	–	Otterpohl

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Table 4.1 Modelling capability included in the three BSMs. A '✓' signifies that the benchmark includes that behaviour or capability, but its implementation is simulation-specific (*Continued*).

	Benchmark model		
	BSM1	**BSM1_LT**	**BSM2**
Secondary sedimentation	Takács	Takács	Takács
Thickener	–	–	ideal mass balance
Dewatering	–	–	ideal mass balance
Reject water storage	–	–	non-reactive tank
Sensors and actuators	✓	✓	✓
Inhibition and toxicity	✓	✓	✓
Temperature impacts	–	✓	✓

NOTE: Space limitation requires that all of the BSM sub-models be summarised only here. In this chapter the reader will find brief model overviews and discussions related to specific model features of interest (new findings, short-comings in the published models, . . .) related to the BSM implementations. This chapter introduces all of the models, but for detailed descriptions of the various models, the reader is referred to the original publications and the detailed technical reports that accompany this STR.

4.1 INFLUENT MODELLING

In any wastewater treatment modelling project, the influent is the single most important feature that needs to be well understood. This is equally true for the benchmark. The influent load impacts sludge production, fractionation of that load impacts the aeration requirements and nutrient removal possibilities and the dynamics in the influent drive the short- and long-term process dynamics. The influent disturbances used to test a particular control strategy play a critical role in the evaluation. That is, because of the multi-faceted nature of the activated sludge process, a particular control strategy may react well to one disturbance and not well to another. Hence, for an unbiased and complete evaluation, it is important that a series of disturbances be defined and that each control strategy be subjected to all the disturbances. Only then can a fair comparison be made.

The influent to the benchmark models has undergone several iterations over the years. BSM1 used a series of 14-day data files, but this approach was insufficient for BSM1_LT and BSM2, so a comprehensive influent model was developed. This section provides only an overview of the different influent options, so the reader is referred to the detailed technical reports or published literature for a complete and detailed description of the approaches.

4.1.1 BSM1 influent

The influent to BSM1 is based on three different 'weather files', corresponding to dry, storm and rain weather disturbance scenarios (Vanhooren & Nguyen, 1996; Spanjers *et al.* 1998; Copp, 1999; Copp, 2002; Copp *et al.* 2002). Each of the files contains 14 days of influent data at 15-minute intervals and defines the influent – in this case, the influent is assumed to be primary effluent – using the ASM1 state variables (Section 4.2.1). In general, these files were developed based on real data and depict expected diurnal variations in influent flow, as well as COD and nitrogen concentrations. The *dry weather* file depicts what is considered to be *normal* diurnal variations. In this file, the resultant peaking factor is 1.74 for maximum

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flow and 2.34 for maximum COD mass load (i.e., flow∗ concentration, mass/day) as compared to the flow-weighted average values. The second file is a variation on the first with the incorporation of two storm events in the second week. The first storm event in this file is a high intensity and short duration event and is expected to 'flush' the sewer of particulate material. The resuspension of these particles is reflected in the data through a significant increase in inert and biodegradable suspended solids. The second storm event assumes the sewers were cleared of particulate matter during the first storm event; hence, only a modest increase in COD load is noted during the second storm. This behaviour is assumed even though the peak flow for both storms is the same and the peak flow of the second storm is maintained over a longer period of time. The third file is meant to represent a long rain event in the second week. The influent flow during this rain event does not reach the level attained during the storm events, but the increased flow is sustained for a much longer period of time. Unlike the storm events, there is no increase in COD load to the plant during the rain event. The flow-weighted average concentrations of the influent components for the three files are shown in Table 4.2.

Table 4.2 Flow-weighted average influent composition in the influent files.

Component	Dry weather	Storm event	Rain event	Units
S_S	69.50	64.93	60.13	g COD · m^{-3}
$X_{B,H}$	28.17	27.25	24.37	g COD · m^{-3}
X_S	202.32	193.32	175.05	g COD · m^{-3}
X_I	51.20	51.92	44.30	g COD · m^{-3}
S_{NH}	31.56	29.48	27.30	g N · m^{-3}
S_I	30.00	28.03	25.96	g N · m^{-3}
S_{ND}	6.95	6.49	6.01	g N · m^{-3}
X_{ND}	10.59	10.24	9.16	g N · m^{-3}
Q	18 446	19 745	21 320	m^3 · day^{-1}

4.1.2 BSM1_LT and BSM2 influent

When the primary clarifier was added to the definition of Benchmark Simulation Model No. 2 (BSM2), it was assumed that the influent to this settler could be adjusted to achieve a primary effluent that was identical to the BSM1 influent. Unfortunately, this assumption proved to be incorrect as it was not possible to achieve the exact BSM1 influent dynamics (equivalent to the BSM2 primary effluent) with a reasonable raw influent to the primary clarifier model. As a result, a separate approach to generate dynamic influent disturbances was adopted for BSM2 and this approach was subsequently used for BSM1_LT as well.

The development of BSM1_LT and BSM2 also highlighted another issue. The evaluation procedure in BSM1 is based on seven (7) days of output data so the use of 28-day influent files was deemed sufficient (two 14-day files back-to-back). However, not all control options can be suitably evaluated within a 7-day period. This is especially true for strategies that involve 'slow' process variables such as solids retention times and/or waste sludge rates. To address this limitation for BSM1_LT and BSM2, the simulation period was increased from 28 days to 18 months (with the last twelve months used for the evaluation) and a new influent modelling tool (Gernaey *et al.* 2006, 2011) was developed to generate the influent for these simulations. This new tool was used to generate influent data for both BSM1_LT and BSM2 even though BSM2 includes a primary clarifier, and BSM1_LT does not. The generic nature of the tool enabled the generation of a primary effluent influent for BSM1_LT and a raw wastewater influent for BSM2.

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Lastly, even though the influents for BSM1_LT and BSM2 are fully defined, it must be understood that the data are based on a model and that the influent model includes typical phenomena that are observed in a year of full-scale WWTP influent data:

- Diurnal behaviour;
- Weekend pattern, which consists of a lower average flow rate and lower pollutant loads during weekends compared to normal weekdays;
- Seasonal changes to model increased infiltration in the rainy season compared to the dry season;
- Holiday periods during which time lower average wastewater flow rates are maintained over a period of several weeks;
- Temperature dynamics;
- Rain events.

The complete model is based on a series of sub-models that, when combined, produce an influent that reflects the various phenomena discussed above. The dry weather model for instance might be combined with rain and storm weather generation to account for 'first flush' effects from the sewer network and dilution phenomena that are typically observed at full-scale WWTPs. An industrial block is included to account for industrial loads and a temperature model is included so that temperature effects also can be described.

Flow rate design principles

The model blocks used to generate the influent flow rate profiles are shown in Figure 4.1. The BSM1 dry weather influent (Q_{av} = 18 446 $m^3 \cdot d^{-1}$; Q_{av} on weekdays = 19 346.3 $m^3 \cdot d^{-1}$; Q_{av} on weekend days = 16 196.3 $m^3 \cdot d^{-1}$) was used as the basis for the generation of the new influent flow profile. Using these BSM1 flows, assumptions were made with respect to the flow contributions from infiltration, households and industrial sources (~25%, ~62% and ~13%, respectively).

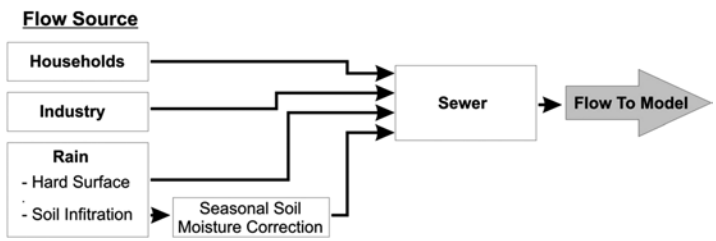

Figure 4.1 General overview of the influent flow rate model.

As illustrated in Figure 4.1, the generation of the influent flow rate is achieved by combining the contributions from households, industry, infiltration and rain with the combination giving the overall sewer flow rate. This flow is then passed through a simple sewer system model (variable volume reservoir series) to achieve the final benchmark influent flow. Each model block that makes up Figure 4.1 is the combination of a number of underlying models and assumptions. As an example, details of the *Households* model block are shown in Figure 4.2.

The basic steps in the flow generation from the *Households* model are described here, but the reader is referred to the detailed technical report and published articles for a more complete description.

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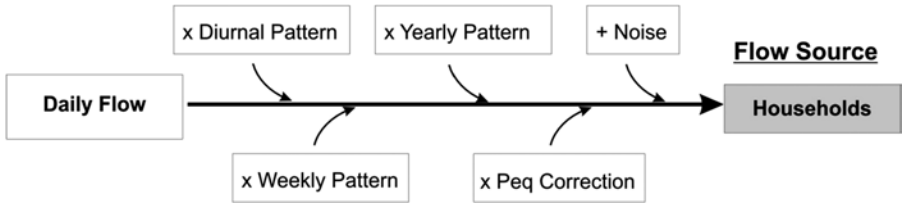

Figure 4.2 *Households* model block, with diurnal flow rate variations, weekend effects and holiday effects.

The basic approach involves a diurnal profile corresponding to the diurnal variation of the wastewater production of one person equivalent (peq) that is normalised and then multiplied by a daily flow rate per person (i.e., 150 $(L \cdot d^{-1}) \cdot peq^{-1}$) to give a dynamic daily flow profile per person. A second function is used to superimpose a weekly household flow rate pattern on the diurnal profiles. In the BSM case, the weekend effect is a slight reduction in the household wastewater production (8% reduction on Saturday, 12% on Sunday) compared to normal weekdays. Similar to the weekend effect, a holiday effect that decreases the flow for three (3) weeks, is added on top. The holiday effect results in a 25% decrease in the flow during the first two weeks, and a 12% decrease during the third holiday week. To increase the realism in the data, white noise, generated when a random number is multiplied by a gain, is added to the calculated flow rate. A sanity check is included to avoid negative values, and to avoid unrealistically high flow rates. Finally, the wastewater production is multiplied by the number of person equivalents in the catchment area to give the estimated household contribution to the total influent flow.

Figure 4.3 illustrates the impact of the various blocks on the calculated diurnal flow rate profiles generated by the *Households* model. Figures 4.3a and 4.3b show the impact of adding noise. Figure 4.3c indicates the impact of the weekend effect and Figure 4.3d shows the effect of adding noise. The holiday effect is presented in Figures 4.3e and 4.3f. These figures show only the average daily flows to minimise the number of data points, but the figures clearly demonstrate a significant reduction in the influent flow rate for a number of days. When tested (without noise), this approach produced an annual average household flow rate equal to 11 513 $m^3 \cdot d^{-1}$, which is very close to 62% of 18 446 $m^3 \cdot d^{-1}$ which was the assumed household contribution in the BSM1 influent flow.

A similar approach is used to calculate the industrial contribution to the influent flow. A weekly pattern including a Friday and weekend effect is combined with a holiday effect that corresponds to the same period as the *Household* holiday period. The *Industry* contribution also includes a one-week holiday effect towards the end of the year. Noise is added and the final result is passed through a sanity check to avoid negative flow rates or rates that are too high.

Rainfall is assumed to contribute to the total flow in two ways: the largest portion of the rainfall flow is assumed to be runoff from impervious surfaces. However, it is also assumed that rainfall on pervious surfaces will influence the groundwater level and contribute to infiltration. The model assumes two seasons and the *Seasonal correction infiltration* model is used to create a dry and wet weather seasonal effect. This seasonal correction is combined with the rainfall and the net contribution of infiltration is calculated. The details related to the calculation of the rainfall portion of the influent flow are described elsewhere (Gernaey *et al.* 2011).

Composition design principles

Similar to the influent flow generation, it is assumed that there are two pollutant sources: households (*Households pollutants* model) and industry (*Industry pollutants* model). The complexity of the model is

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reduced significantly by neglecting all other influent pollutant sources (infiltration and rain). Pollutant fluxes from both sources are calculated and combined and then the particulate concentrations are altered based on flow to account for first flush effects. The resulting influent vector, consisting of the 13 ASM1 state variables, the TSS concentration (as defined in Chapter 6), a flow rate and temperature, can be used directly as an input to any ASM1 compatible simulation model. The BSM pollutant loads were designed based on the assumption that 20% of the COD load and 10% of the nitrogen load are due to industrial activity in the BSM1 influent.

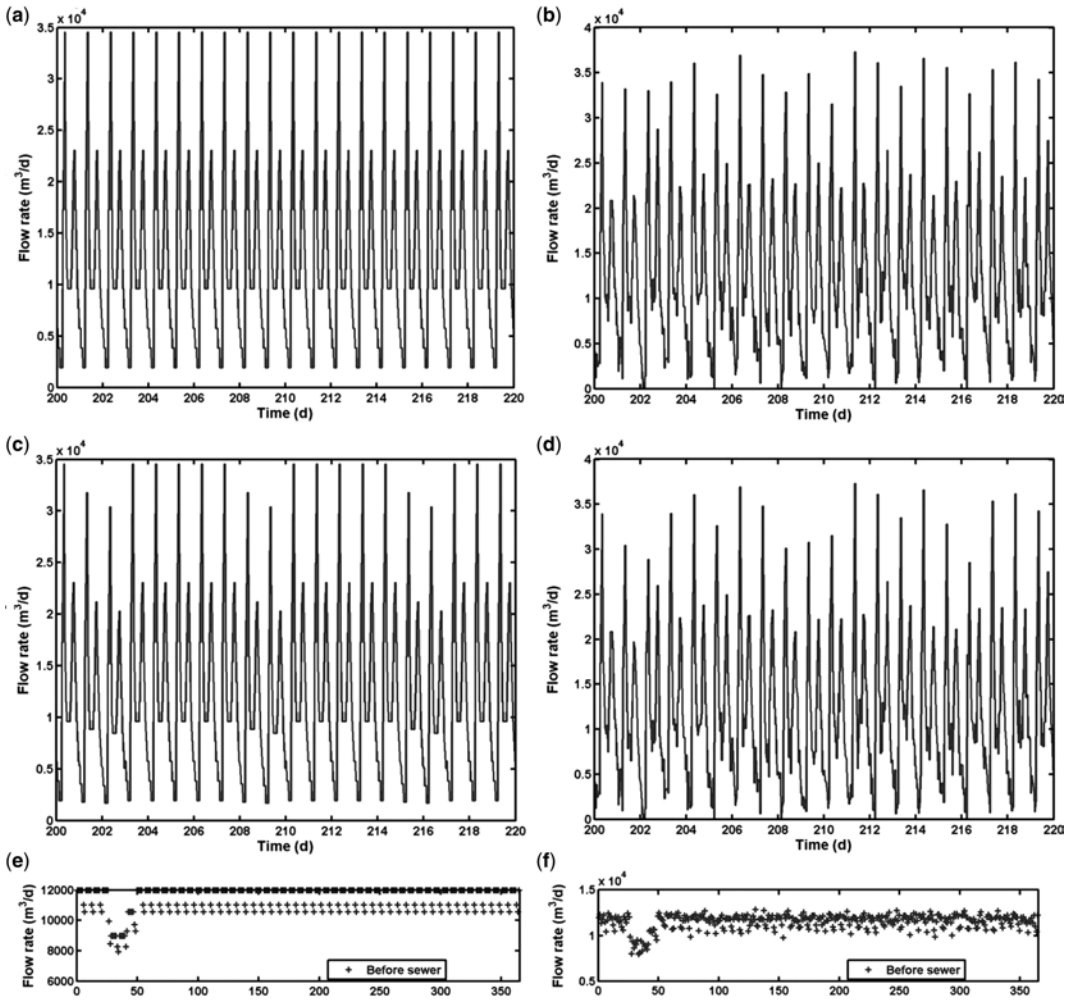

Figure 4.3 Dynamic flow rate profiles (15 min. sampling interval) resulting from the *Households* model block: (a) Only diurnal effect, no noise added; (b) As (a), but with noise added in the *Households* model block; (c) Diurnal effect combined with weekend effect, no noise added; (d) As (c), but with noise added; (e) Daily average flow rates (each average daily flow rate value represents the mean of 96 samples), combining diurnal, weekend and holiday effects, no noise added; (f) As (e), but with noise added. Note that the first day in (e) and (f) corresponds to July 1st.

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A generalised approach, based on particulate COD (COD_{part}), soluble COD (COD_{sol}), ammonium (S_{NH}) and Kjeldahl nitrogen (TKN) has been adopted for the composition of the influent (Figure 4.4). The advantage of this approach is that these components can be used for generating influent files for other ASM models making this tool universally applicable and not simply a benchmark model (Gernaey *et al.* 2011).

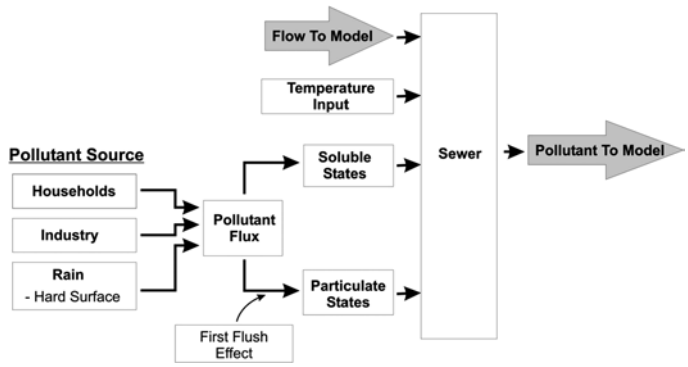

Figure 4.4 Overview of the modules in the proposed model for influent data generation.

Diurnal profiles (24 values per day) for soluble and particulate COD (COD_{sol}, COD_{part}), ammonium (S_{NH}) and Kjeldahl nitrogen (TKN) fluxes form the basis for the dynamic profiles generated with the *Households pollutants* model (Figure 4.5). The values in the four input files are normalised and the pollutant fluxes are transformed into $(g\ COD \cdot peq^{-1}) \cdot d^{-1}$ and $(g\ N \cdot peq^{-1}) \cdot d^{-1}$ units by multiplying the values in the input files by specific scaling constants (i.e., a different constant for each profile) and then by a constant representing the number of person equivalents in the catchment. These mass load profiles are then subjected to transformations that account for the weekend and holiday effects. The weekend effect corresponds to a reduction of the household pollutant fluxes (12% reduction on Saturday, 16% on Sunday) and the holiday effect uses the same reduction approach as for the flow rates. Uncorrelated white noise is added to each pollutant flux and a sanity check is applied to avoid negative values.

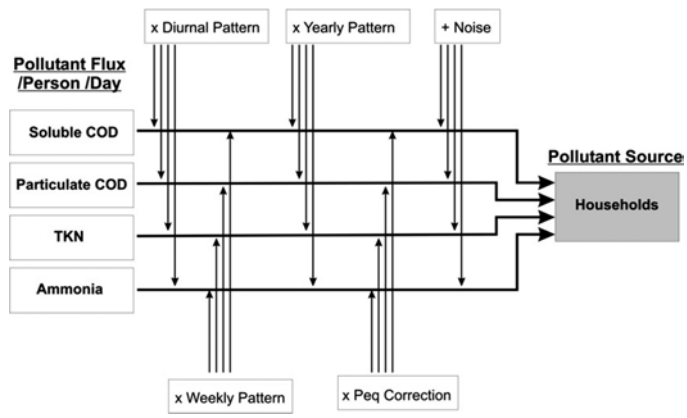

Figure 4.5 Layout of the *Households pollutants* model.

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The *Industry pollutants* model (Figure 4.6) has a similar structure. Weekly profiles for COD_{sol}, COD_{part}, S_{NH} and TKN that contain diurnal pollutant flux variations, a Friday afternoon pollutant flux peak load and a weekend effect form the basis for the industrial pollutant fluxes. A holiday effect is added using the same input approach as for the industrial flow. After multiplying the weekly flux pattern with the holiday pattern, zero mean white noise is added and a sanity check is applied to avoid negative values.

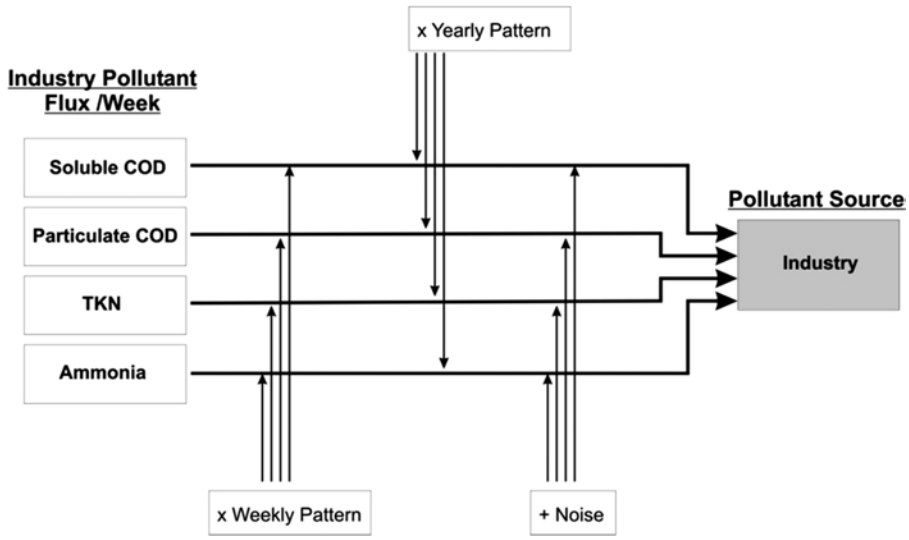

Figure 4.6 Layout of the *Industry pollutants* model.

With the loads determined, an *ASM1_fractionator* model converts the bulk pollutant concentrations to state variable concentrations that are compatible with the ASM1 model (Table 4.3). Several basic principles are used to generate the state variables. For instance, the following rules were used:

- Provided that the COD_{sol} flux is greater than 30 g COD · m^{-3}, the S_I concentration is assumed to be constant and equal to 30 g COD · m^{-3}. It is furthermore assumed that S_I is present in the dry weather wastewater flow. Only rainfall runoff from impervious areas can dilute S_I.
- The S_S concentration is calculated as the difference between the COD_{sol} flux and the S_I pollutant flux.
- $X_{B,A}$ and X_P are assumed to be zero.
- The COD_{part} is distributed between X_I, X_S and $X_{B,H}$ according to the BSM1 flow weighted average dry weather influent (18.2%, 71.8% and 10.0%, respectively).
- The S_O (oxygen) is assumed to be zero.
- The S_{NO} (nitrate) is zero.
- The S_{NH} concentration is calculated based on the S_{NH} flux.
- The organic N flux is obtained by subtracting the S_{NH} flux and the N flux associated with the organic particulate variables, $X_{B,H}$ and X_I. If the result of this calculation is greater than zero, the remaining organic N is distributed between S_{ND} and X_{ND} based on the S_{ND}/X_{ND} ratio in the BSM1 influent files (39.6% and 60.4%, respectively). If the result of this calculation is negative, influent S_{ND} and X_{ND} are assumed to be zero.

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Table 4.3 Average flow-weighted influent concentrations calculated for one year of influent data generated with the BSM2 influent model.

Variable	Concentration	Unit
$S_{I,inf}$	27.21	$g\ COD \cdot m^{-3}$
$S_{S,inf}$	58.15	$g\ COD \cdot m^{-3}$
$X_{I,inf}$	92.46	$g\ COD \cdot m^{-3}$
$X_{S,inf}$	363.77	$g\ COD \cdot m^{-3}$
$X_{B,H,inf}$	50.66	$g\ COD \cdot m^{-3}$
$X_{B,A,inf}$	0.00	$g\ COD \cdot m^{-3}$
$X_{P,inf}$	0.00	$g\ COD \cdot m^{-3}$
$S_{O,inf}$	0.00	$g\ O_2 \cdot m^{-3}$
$S_{NO,inf}$	0.00	$g\ N \cdot m^{-3}$
$S_{NH,inf}$	23.85	$g\ N \cdot m^{-3}$
$S_{ND,inf}$	5.64	$g\ N \cdot m^{-3}$
$X_{ND,inf}$	16.12	$g\ N \cdot m^{-3}$
$S_{ALK,inf}$	7.00	$mol\ HCO_3^- \cdot m^{-3}$
TSS_{inf}	380.17	$g \cdot m^{-3}$
Q_{inf}	20 668.44	$m^3 \cdot d^{-1}$
T_{inf}	14.86	°C

4.2 UNIT PROCESS MODELS

To increase the acceptability of the results, four internationally accepted process models were chosen. Activated Sludge Model No. 1 (ASM1) was chosen as the biological process model (Henze *et al.* 1987), Anaerobic Digestion Model No. 1 (ADM1) was chosen as the anaerobic digestion model (Batstone *et al.* 2002), the Otterpohl (1995) model was chosen for primary settling and the double-exponential settling velocity function of Takács *et al.* (1991) was chosen as a fair representation of the secondary settling process.

4.2.1 Activated Sludge Model No. 1 (ASM1)

For the biological reactors, ASM1 was chosen for its universal appeal. It is acknowledged however, that there are several limitations with ASM1, and several activated sludge models have been developed since ASM1 was first introduced, including ASM2d and ASM3. Nevertheless, its practical validation and the international acceptance of ASM1 at the time the first benchmark was developed was the overriding criterion for its use. A complete description of the model and its development are available elsewhere (Henze *et al.* 1987). Table 4.4 lists the ASM1 state variables, the associated symbols and the state variable units.

To ensure the consistent application of the model in all benchmarking studies, all of the kinetic and stoichiometric model parameters have been defined in the simulation benchmark description (Appendix A). Included in these tables are the parameter descriptions, their recognised symbols

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and units as well as an associated value. The listed parameter estimates approximate those that are expected at 15°C.

Table 4.4 State variables for the IWA Activated Sludge Model No. 1 (ASM1).

State variable description	State symbol	Units
Soluble inert organic matter	S_I	g COD · m^{-3}
Readily biodegradable substrate	S_S	g COD · m^{-3}
Particulate inert organic matter	X_I	g COD · m^{-3}
Slowly biodegradable substrate	X_S	g COD · m^{-3}
Active heterotrophic biomass	$X_{B,H}$	g COD · m^{-3}
Active autotrophic biomass	$X_{B,A}$	g COD · m^{-3}
Particulate products arising from biomass decay	X_P	g COD · m^{-3}
Dissolved oxygen	S_O	g O_2 · m^{-3}
Nitrate and nitrite nitrogen	S_{NO}	g N · m^{-3}
$NH_4^+ + NH_3$ nitrogen	S_{NH}	g N · m^{-3}
Soluble biodegradable organic nitrogen	S_{ND}	g N · m^{-3}
Particulate biodegradable organic nitrogen	X_{ND}	g N · m^{-3}
Alkalinity	S_{ALK}	mol HCO_3^- · m^{-3}

4.2.2 Anaerobic Digestion Model No. 1 (ADM1)

As with ASM1, the choice of ADM1 as the anaerobic digestion model was driven by its availability, potential for integration and international exposure. All of the models in the BSMs were thoroughly tested and this provided a great deal of insight into the model description as well as the model implementations in various platforms. The introduction of ADM1 into BSM2 was no exception. The BSM2 implementation identified several issues not explicitly discussed in the published model. The description here addresses some of those issues. It should be noted that the ADM1 implementation for BSM2 deviates somewhat from the model description in Batstone *et al.* (2002) and there are three main reasons for this. Firstly, ADM1 has been implemented so that it is consistent with the full BSM2. Secondly, computational issues have been addressed (Rosen *et al.* 2006) and lastly, unpublished parameter values have been chosen in consultation with the original ADM model developers (Batstone 2002–2008, personal communication). Readers are referred to the detailed technical report and original publication for a complete description of ADM1.

4.2.2.1 Elemental balances

Maintaining a complete elemental balance for all model components (COD, N, . . .) is a fundamental issue for any model and was explored in detail in Hauduc *et al.* (2010) where the authors attempted to verify several published activated sludge models. However, the implementation of ADM1 into the benchmark identified a few potential problems.

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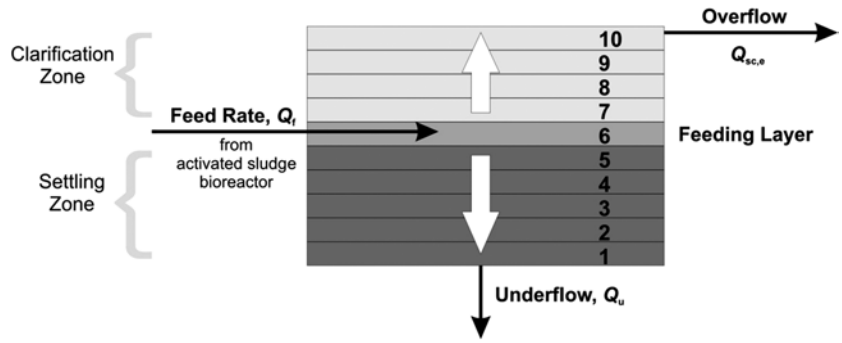

Figure 4.10 Schematic model of the secondary clarifier.

The double-exponential settling velocity function that is used for determining the particle settling velocities inside each layer uses a number of parameters and these have been fully defined in Appendix A. The general equation is:

$$v_{sj} = v_o e^{-r_h X_j^*} - v_o e^{-r_p X_j^*}$$
$$0 \le v_{sj} \le v_o'$$

where v_{sj} is the settling velocity in layer j [m · day^{-1}], X_j^* is the solids concentration in layer j [g · m^{-3}], subject to the limiting condition $X_j^* = X_j - X_{min}$, X_j is the suspended solids concentration in layer j [g · m^{-3}], and X_{min} is the minimum attainable suspended solids concentration [g · m^{-3}] calculated as $X_{min} = f_{ns} \cdot X_{in}$ (X_{in} is the mixed liquor suspended solids concentration entering the settling tank and f_{ns} is the non-settleable fraction). Figure 4.11 shows the settling velocity curve that is obtained with the benchmark parameter values.

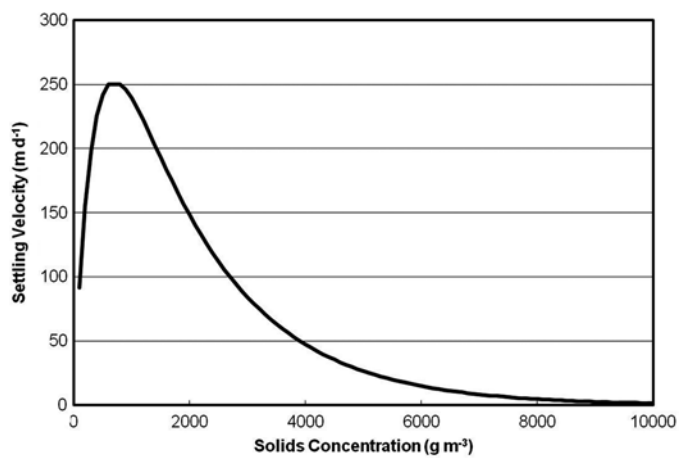

Figure 4.11 Settling velocity curve obtained using the benchmark parameter values.

As the benchmark implementation assumes a non-reactive model, the mass balances are applied to the calculated incoming total suspended solids (TSS) and not to the individual particulate ASM state variables.

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The oxygen transfer coefficient, K_La, is also dependent on temperature. ASCE (1993) presents the generally accepted dependency between the oxygen transfer coefficient K_La and temperature:

$K_La(T) = 1.024^{(T-15)} \cdot K_La_{15}$ with K_La in d^{-1} and T in °C.

This equation must be interpreted as follows: for a given energy input (or equivalently, a certain gas flow rate), which would yield a K_La of 240 d^{-1} at 15°C, the oxygen transfer efficiency drops to approximately 213 d^{-1} at 10°C, and increases to 270 d^{-1} when the temperature in the mixed liquor increases to 20°C.

It was decided that the aeration control handle should continue to be based on $K_La(15)$ with the modelled $K_La(T)$ being calculated based on the current model temperature. What this means is that to reach a certain oxygen concentration a higher $K_La(15)$ will be required if the reactor temperature is low and a lower value when the temperature is high. This leads to the known phenomenon that more energy will be needed for aeration at low temperatures compared to high temperatures. Fortunately, the variation in the oxygen saturation concentration with temperature compensates for this reduced mass transfer efficiency, so that overall, the energy needed to reach a certain oxygen concentration is not affected all that much by temperature (i.e., thanks to the increased driving force at low temperatures, a lower $K_La(15)$ will do the job).

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Chapter 5

Benchmarking of control strategies

K. V. Gernaey, J. B. Copp, U. Jeppsson, I. Nopens, M.-N. Pons and P. A. Vanrolleghem

The main purpose of the benchmark models is the simulation-based comparison of control and process monitoring strategies. However, in practice the major part of this Scientific and Technical Report is focused on the modelling aspects of wastewater treatment plants. This illustrates that careful selection of each unit process model, its implementation and ring-testing, the fine tuning of its parameters and finally its approval as part of the benchmark models has been a valuable but also time-consuming exercise. Here, the Task Group would like to add explicitly that no realistic evaluation of control strategies would have been possible without these modelling efforts. Thus, in order to highlight the fundamental purpose of the benchmark models, this chapter focuses on benchmarking of control strategies. More specifically, certain aspects of controllers for BSM1, BSM1_LT and BSM2 will be highlighted.

5.1 BSM1 AND BSM1_LT CONTROLLERS

5.1.1 Default BSM1 control strategy

Default controllers are proposed for BSM1, such that the closed-loop simulation and the implementation of the evaluation criteria (Chapter 6) can be verified using the default strategy before the user implements his/her own control strategy. In this way the user is given an additional opportunity – in addition to the steady state and open-loop dynamic simulation results – to ensure that their BSM1 implementation is functioning properly. The primary control objectives for the default strategy in BSM1 are: (i) to maintain the NO_3-N concentration in the second anoxic tank at a predetermined set point value (1 g N · m^{-3}); and, (ii) to maintain the dissolved oxygen concentration in the fifth tank, that is, the last aeration tank, at a predetermined set point value (2 g O_2 · m^{-3}). Both default controllers are Proportional-Integral (PI) controllers. It should be emphasised that this defined control strategy is not the best strategy available. It is only provided as a test case for the user.

In this case, the NO_3-N sensor in the second anoxic compartment is a class B_0 sensor with a measurement range from 0 to 20 g N · m^{-3}. An explanation of the BSM sensor classes can be found in Chapter 4. The measurement noise is equal to 0.5 g N · m^{-3}. The manipulated variable is the internal recycle flow rate from the last aerated tank back to the first tank (Q_{int}).

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For the dissolved oxygen control in the last aerated compartment, the probe is assumed to be a class A sensor with a measurement range from 0 to 10 g $O_2 \cdot m^{-3}$ and measurement noise of 0.25 g $O_2 \cdot m^{-3}$. The manipulated variable is the oxygen transfer coefficient for the last tank, K_La_5.

To increase the degree of realism in the simulations, constraints on the manipulated variables are applied. The internal recirculation flow (Q_{int}) is constrained to between 0 and 5 times the average dry weather influent flow rate to the plant (18 446 $m^3 \cdot d^{-1}$). The external recycle flow rate (Q_r) is held constant at the average dry weather influent flow rate to the plant. For the oxygen transfer in tank 5, it is assumed that K_La can only vary from 0 to 360 d^{-1}. Furthermore, whenever the K_La in an aerated tank is below 20 d^{-1}, it is assumed that mechanical mixing is needed in order to keep the activated sludge in suspension.

5.1.2 Other BSM1 control handles

All available control handles, except aeration (see below), are considered to be ideal. In the closed-loop test case, only two control handles are used: the internal recirculation flow rate (Q_{int}) and the oxygen transfer rate in tank number 5 (K_La_5). However, the following control handles are available for the implementation of user-defined control strategies in BSM1:

- return sludge flow rate (Q_r);
- anoxic/aerobic volume – all five biological tanks are assumed to be equipped with both aerators and mechanical mixing devices, that is, in a discrete fashion the volumes for anoxic and aerobic behaviour can be modified;
- aeration intensity individually for each tank (K_La_1, K_La_2, K_La_3, K_La_4, K_La_5), taking into account the dynamics of the aeration system;
- external carbon source flow rate (Q_{EC1}, Q_{EC2}, Q_{EC3}, Q_{EC4}, Q_{EC5}) where the carbon source is considered to consist of readily biodegradable substrate, that is, $COD_{EC} = S_S$;
- influent distribution by use of step feed (fractions of the influent flow to each of the five biological tanks: $f_{Qi1}, f_{Qi2}, f_{Qi3}, f_{Qi4}, f_{Qi5}$);
- distribution of internal flow recirculation (fractions of the internal recirculation flow to each of the five biological tanks: $f_{Qint1}, f_{Qint2}, f_{Qint3}, f_{Qint4}, f_{Qint5}$) – in principle internal flow recirculation between any of the tanks is allowed;
- distribution of return sludge flow (fractions of the return sludge flow to each of the five biological tanks: $f_{Qr1}, f_{Qr2}, f_{Qr3}, f_{Qr4}, f_{Qr5}$).

The non-ideal aeration system (K_La_1- K_La_5) is defined with significant dynamics and a response time of $t_r = 4$ min (see Rieger *et al.* 2006). A second-order time delay function gives a reasonable model of this process. The time constant of each of the two identical first-order delays is $\tau = t_r/3.89 = 1.03$ min.

The above selection of control handles dramatically increases the flexibility of the defined BSM1 system. This number of control handles may not be realistic for a real plant but they have been included here in order to encourage all types of control strategy evaluations.

5.1.3 BSM1_LT control strategy

Many control actions at a wastewater treatment plant act on the process over longer time scales and can thus not be evaluated using BSM1 with its one-week evaluation period. For example, a short-term evaluation does not allow for realistic equipment (sensor/actuator) modelling, including failures, drift and maintenance, as the impact of these phenomena typically appear over longer time scales. In order to cope with these limitations of BSM1, BSM1_LT was proposed and developed (Rosen *et al.* 2004; Corominas *et al.* 2011).

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Chapter 6

Evaluation criteria

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The definition of the BSM simulation procedure discussed in the next chapter is meant to ensure that any data analysed by simulation benchmark users is generated in a similar manner. However, the results of these dynamic simulations lead to the question of how the huge amount of output data is to be evaluated. To aid the evaluation process, a series of quantities have been identified. These evaluation criteria provide a simple, accurate and objective means for comparison and have been developed specifically for comparing dynamic responses and the impact of different control strategies.

The evaluation criteria, as a whole, are a set of geographically independent measures that combine the output data into a small number of composite terms. The criteria discussed here update and extend the criteria defined for BSM1 (Copp, 2002; Pons *et al.* 2009), to include the new processes included in BSM2. These composite terms include, among others, a general effluent quality measure, an operational cost index, energy terms for pumping and aeration, cost for external carbon addition, biogas production and a measure of sludge production. The methods used to calculate these terms are outlined below.

To exemplify the evaluation criteria, a complete evaluation example of the open-loop BSM2 is presented in Appendix A. The open-loop BSM2 case is taken directly from Nopens *et al.* (2010) and was simulated exactly according to the description given there. The influent data file used is the one defined by Gernaey *et al.* (2005, 2006, Technical Report No. 8) and available for download at www.benchmarkwwtp.org.

6.1 EFFLUENT AND INFLUENT QUALITY INDICES

The effluent quality index (EQI, in kg pollution units $\cdot$ d^{-1}) is a weighted average sum of relevant effluent concentrations (Copp, 2002). The averaged values used for all evaluation criteria for all BSMs are based on dynamic output values recorded every 15 minutes and assume constant values between samples (zero-order hold). However, the evaluation varies slightly depending on the benchmark system being used. For example, with BSM1, the evaluation period (t_{obs}) is one week (t_{start} = 21 days, t_{end} = 28 days) but in the case of BSM2 and BSM1_LT, the evaluation period (t_{obs}) has been extended to 364 days (t_{start} = 245 days, t_{end} = 609 days). Secondly, whereas there is only one effluent in BSM1 and BSM1_LT (Figure 3.1), the effluent that is used

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to calculate the EQI for BSM2 (Figure 3.2) consists of the combination of the settler effluent, the effluent that bypassed the plant entirely and the effluent that bypassed the activated sludge system only (i.e., this last flow has passed through the primary settler). In BSM2, those three effluent sources are added together and the sum (Q_e in the equation below) is used for the evaluation of the effluent criteria:

$$\mathrm{EQI} = \frac{1}{t_{\mathrm{obs}}1000}\int_{t_{\mathrm{start}}}^{t_{\mathrm{end}}} \Big(\beta_{\mathrm{SS}}\cdot \mathrm{TSS}_e(t) + \beta_{\mathrm{COD}}\cdot \mathrm{COD}_e(t) + \beta_{\mathrm{TKN}}\cdot \mathrm{TKN}(t) + \beta_{\mathrm{NO}}\cdot S_{\mathrm{NO},e}(t) + \beta_{\mathrm{BOD5}}\cdot \mathrm{BOD}_{5,e}(t)\Big)\cdot Q_e(t)\cdot dt$$

The definitions for the composite variables are similar for all the systems (Copp, 2002) and the weighting factors (β_x) for the different concentrations are based on Vanrolleghem *et al.* (1996) with the TKN and S_{NO} factors modified to promote ammonia removal. The values for the weighting factors are: $\beta_{\mathrm{SS}} = 2$, $\beta_{\mathrm{COD}} = 1$, $\beta_{\mathrm{TKN}} = 30$, $\beta_{\mathrm{NO}} = 10$ and $\beta_{\mathrm{BOD5}} = 2$.

$$\mathrm{TSS} = 0.75\cdot(X_{\mathrm{S}} + X_{\mathrm{I}} + X_{\mathrm{B,H}} + X_{\mathrm{B,A}} + X_{\mathrm{P}})$$
$$\mathrm{COD} = S_{\mathrm{S}} + S_{\mathrm{I}} + X_{\mathrm{S}} + X_{\mathrm{I}} + X_{\mathrm{B,H}} + X_{\mathrm{B,A}} + X_{\mathrm{P}}$$
$$\mathrm{TKN} = S_{\mathrm{NH}} + S_{\mathrm{ND}} + X_{\mathrm{ND}} + i_{\mathrm{XB}}\cdot(X_{\mathrm{B,H}} + X_{\mathrm{B,A}}) + i_{\mathrm{XP}}\cdot(X_{\mathrm{P}} + X_{\mathrm{I}})$$

It should be noted that ASM1 does not include a state variable for inorganic particulate matter so a completely realistic and dynamic TSS value cannot be modelled with ASM1. This recognised shortcoming of ASM1 was compensated for by using a value of 0.75 to relate particulate COD to TSS. This approach is not ideal as it results in a perfect correlation between the inorganic particulate material and the total particulate COD content, but it is an attempt to acknowledge that inorganic material will exist in the system.

The definitions for TSS ($\mathrm{g\cdot m^{-3}}$), COD ($\mathrm{g\ COD\cdot m^{-3}}$) and TKN ($\mathrm{g\ N\cdot m^{-3}}$) are the same for all streams, but the definition for $\mathrm{BOD_5}$ ($\mathrm{g\ BOD_5\cdot m^{-3}}$) is not. The benchmark system assumes that $\mathrm{BOD_5}$ in all streams is a function of the degradable organic state variables. However, different factors are applied to each stream depending on the source of that material. The assumption is made that the influent contains more readily biodegradable material and a factor of 0.65 is applied to the influent organic content. This contrasts the 0.25 factor used for the fully processed secondary effluent, which typically contains less readily biodegradable material and more biomass and slowly biodegradable organic matter. Of course all effluent that has bypassed the activated sludge system uses the influent factor, 0.65. As a consequence, the effluent $\mathrm{BOD_5}$ is calculated as:

$$\mathrm{BOD}_{5,\mathrm{e}} = (\mathrm{BOD}_{5,\mathrm{e,s}}\cdot Q_{\mathrm{e,s}} + \mathrm{BOD}_{5,\mathrm{e,bp}}\cdot Q_{\mathrm{e,bp}} + \mathrm{BOD}_{5,\mathrm{e,bas}}\cdot Q_{\mathrm{e,bas}})/Q_{\mathrm{e}}$$

where,

$$\mathrm{BOD}_{5,\mathrm{e,s}} = 0.25\cdot(S_{\mathrm{S,s}} + X_{\mathrm{S,s}} + (1 - f_{\mathrm{P}}')\cdot(X_{\mathrm{B,H,s}} + X_{\mathrm{B,A,s}}))$$
$$\mathrm{BOD}_{5,\mathrm{e,bp}} = 0.65\cdot(S_{\mathrm{S,bp}} + X_{\mathrm{S,bp}} + (1 - f_{\mathrm{P}}')\cdot(X_{\mathrm{B,H,bp}} + X_{\mathrm{B,A,bp}}))$$
$$\mathrm{BOD}_{5,\mathrm{e,bas}} = 0.65\cdot(S_{\mathrm{S,bas}} + X_{\mathrm{S,bas}} + (1 - f_{\mathrm{P}}')\cdot(X_{\mathrm{B,H,bas}} + X_{\mathrm{B,A,bas}}))$$

with the subscripts ‘s’, ‘bp’ and ‘bas’ referring to the settler effluent, effluent that bypassed the entire plant and the effluent that bypassed the activated sludge system, respectively.

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Note: Thanks to the rigorous attention of one of the STR reviewers, the authors acknowledge that an error in these BOD equations has existed since they were first introduced nearly 20 years ago. Based on this review, the above equations now refer to an f_p' term (rather than f_p as previously used) and that constant should have a value of 0.20. The Task Group believes that this value is more consistent with modelling practice and differentiates this constant from the ASM1 model parameter, f_p. Two things should be further noted: i) although the parameter value change will have a minimal impact on the BOD calculations, it was deemed important to make this change; and, ii) all BOD results reported in the STR were calculated with the incorrect value, but going forward BOD values will be calculated using the corrected value of 0.20.

Besides the effluent quality, the influent quality can be calculated for reference purposes. The influent quality index (IQI, in kg pollution units $\cdot$ d^{-1}) is calculated in the same way as the effluent quality, as a weighted average sum of corresponding influent concentrations, with BOD_5 calculation using the 0.65 factor.

6.2 EFFLUENT CONCENTRATIONS

Specific components in the effluent are evaluated using three quantities including 95 percentile and violation reporting. Effluent limit violations are included so that the control strategies are compared not only based on their ability to meet a control objective, but also on their ability to provide effective treatment. The effluent violations are calculated from the output data generated at 15-minute intervals and are reported through two quantities: (i) number of violations; and, (ii) percentage of time the plant is in violation. Constraints with respect to five effluent components are defined and the percentage of time that the constraints are not met is to be reported. As well, the methodology for reporting the number of violations is defined. The same effluent limits are defined for all of the benchmark models:

- $N_{tot,e} < 18$ g N $\cdot$ m^{-3},
- $COD_e < 100$ g COD $\cdot$ m^{-3},
- $S_{NH,e} < 4$ g N $\cdot$ m^{-3},
- $TSS_e < 30$ g TSS $\cdot$ m^{-3} and,
- $BOD_{5,e} < 10$ g $BOD_5 \cdot$ m^{-3}.

The above effluent limits were arbitrarily chosen and do not correspond to any effluent requirements in any known location. For the purposes of the benchmark, it should be noted that these limits are compared to the instantaneous model output generated at 15 minute intervals and are not related to daily, weekly, monthly, quarterly or yearly averages as might be the case with real regulations in a given location.

6.2.1 Ninety-five (95) percentiles

The 95 percentiles are defined as the effluent concentrations that are exceeded only 5% of the time. Percentile concentrations for effluent ammonia ($S_{NH,e95}$), effluent total nitrogen ($N_{tot,e95}$) and total suspended solids (TSS_{e95}) are to be reported as part of the evaluation criteria.

6.2.2 Number of violations

This quantity represents the number of times that the plant is in violation of the effluent limits (i.e., the number of times the plant effluent concentration increases above the effluent limit during the evaluation

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period). This measure does *not* reflect the length of time that the plant is in violation. An illustrative example from Copp *et al.* (2002) is given below.

6.2.3 Percentage of time plant is in violation

This quantity is a measure of the percentage of the time that the plant is in violation of the effluent limits. An illustrative example is given below.

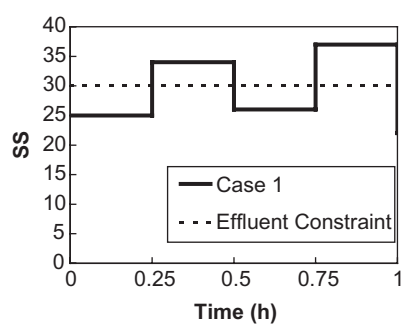
Calculation Example:
(effluent constraint: 30 g · m^{-3})

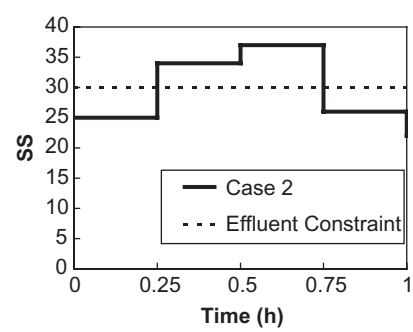
time (h)	Effluent Suspended Solids (g SS · m^{-3})	
	Case 1	Case 2
0.00	25	25
0.25	34	34
0.50	26	37
0.75	37	26
1.00	22	22

Two hypothetical data sets for effluent suspended solids are shown in the table above and depicted in the figures below.

In case 1, the number of violations is 2. That is, the effluent suspended solids rose above the effluent constraint twice in this 1-hour period. During that hour, the effluent limit was in violation for 30 minutes (square wave concentrations are assumed, see figures); hence the percentage of time in violation is 50.

In case 2, the number of violations is 1. That is, the effluent suspended solids rose above the effluent constraint only once in this hour. The fact that it remained above the constraint for an extended period is irrelevant for this quantity. As in case 1, the effluent limit was in violation for 30 minutes so the percentage of time in violation is still 50 for case 2.





6.3 OPERATIONAL COST INDEX

The operational cost index (OCI) is calculated as a weighted sum of different costs:

$$\mathrm{OCI} = AE + PE + f_{\mathrm{sp}} \cdot SP + f_{\mathrm{ec}} \cdot EC + ME - f_{\mathrm{mp}} \cdot MP + \max(0, HE - 7MP)$$

where, AE (kWh · d^{-1}) is aeration energy, PE (kWh · d^{-1}) is pumping energy, SP (kg SS · d^{-1}) is sludge production for disposal, EC is external carbon addition (kg COD · d^{-1}), ME (kWh · d^{-1}) is mixing

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energy, MP (kg $CH_4 \cdot d^{-1}$) is methane production and HE (kWh $\cdot d^{-1}$) is the heating energy needed to increase the temperature of the sludge in the anaerobic digester. The weighting factors are listed in Table 6.1.

Table 6.1 OCI weighting factors.

Factor	BSM1	BSM_LT	BSM2
f_{sp}	5	5	3
f_{EC}	3	3	3
f_{mp}	n/a	n/a	6

The value of f_{sp} was reduced for BSM2 to account for the sludge processing being carried out in that system and the expected sludge disposal cost savings as a result. The weight for EC, f_{EC}, is 3 and the value for f_{mp} is set to 6, which is consistent with the assumption that 43% of the theoretical energy content of the produced methane can be made available as electricity by the assumed gas motor. This also assumes that approximately 50% of the energy is used for heating the digester influent and 7% is lost.

The resulting OCI is dimensionless, but is a quantity that can be used to compare the relative costs between different control strategies.

6.3.1 Aeration energy

In the original BSM1 definition (Copp, 2002), the aeration energy (AE) calculation was based on the following equation with an output in units kWh $\cdot d^{-1}$:

$$AE = \frac{24}{t_{obs}} \int_{t_{start}}^{t_{end}} \sum_{i=1}^{i=5} (3.0248 \cdot 10^{-4} \cdot V_i \cdot K_L a_{i,15}(t)^2 + 0.5882 \cdot 10^{-2} \cdot V_i \cdot K_L a_{i,15}(t)) \cdot dt$$

where $K_L a_{15}$ was the $K_L a$ value calculated at a temperature of 15°C, $K_L a$ is expressed in h^{-1}, V_i is the volume of reactor i (m^3) and t_{obs} represents the total length of the evaluation period in days (i.e., $t_{end} - t_{start}$). Use of this equation demonstrated that, while the temperature effects on $K_L a$ and S_O^{sat} are considered properly, the calculated aeration energy costs tended to dominate the other operating cost quantities such as sludge production. As a result of these findings, a new approach for calculating the aeration cost was developed. This new approach was based on a typical engineering approach related to the SOTE (Standard Oxygen Transfer Efficiency). It was assumed in this approach that at 15°C and a dissolved oxygen concentration of 0 g $O_2 \cdot m^{-3}$, 1.8 kg O_2 can be transferred per kWh under process conditions. This allows AE as a function of $K_L a_{15}$ to be written down explicitly. The amount of oxygen transferred per unit time or more commonly referred to as the oxygen transfer rate (OTR), calculated in units kg $O_2 \cdot d^{-1}$ in a volume V, is given by:

$$\text{OTR} = V \cdot K_L a_{15} \cdot (S_{O,15}^{sat} - 0)/1000$$

where, S_O^{sat} is expressed in g $O_2 \cdot m^{-3}$, V in m^3 and $K_L a_{15}$ in d^{-1}. For such an OTR, the energy required is given by:

$$AE = \frac{S_{O,15}^{sat}}{t_{obs} \cdot 1.8 \cdot 1000} \int_{t_{start}}^{t_{end}} V \cdot K_L a_{15}(t) dt$$

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Because the calculated AE is independent of the dissolved oxygen concentration (i.e., any $K_L a_{15}$ comes with an energy expenditure equal to the one that would lead to 1.8 kg $O_2 \cdot (kWh)^{-1}$ transferred when the DO was zero), the equation can be applied at all times, irrespective of the actual DO concentration. This leads to the new AE function:

$$AE = \frac{S_{O,15}^{sat}}{t_{obs} \cdot 1.8 \cdot 1000} \int_{t_{start}}^{t_{end}} \sum_{i=1}^{i=5} V_i \cdot K_L a_{i,15}(t) dt$$

where, t_{obs} is the period of evaluation (d), V_i is the volume of reactor i (m^3), $K_L a_{i,15}$ is the mass transfer coefficient in reactor i at 15°C (d^{-1}) and $S_{O,15}^{sat}$ is the saturated oxygen concentration at 15°C (i.e., 8 g $O_2 \cdot m^{-3}$). The modified AE calculation is now used in all BSMs.

6.3.2 Pumping energy

Pumping energy ($kWh \cdot d^{-1}$) is calculated as a weighted average sum of the various pumped flows in the system under study. In BSM1, this includes only the external recycle flow Q_r, the internal recycle flow Q_{int} and the waste sludge flow Q_w. In the original BSM1 definition a common weight of 0.04 $kWh \cdot m^{-3}$ was defined for all pumped flows, but this assumption was later modified to include separate weighting factors for each flow. In BSM2, the calculation also includes the primary clarifier underflow Q_{pu}, the thickener unit underflow Q_{tu} and the dewatering unit overflow flow Q_{do} to yield:

$$PE = \frac{1}{t_{obs}} \int_{t_{start}}^{t_{end}} \begin{pmatrix} f_{PE_Q_{int}} \cdot Q_{int} + f_{PE_Q_r} \cdot Q_r + f_{PE_Q_w} \cdot Q_w + \\ f_{PE_Q_{pu}} \cdot Q_{pu} + f_{PE_Q_{tu}} \cdot Q_{tu} + f_{PE_Q_{do}} \cdot Q_{do} \end{pmatrix} dt$$

where all flows are expressed in $m^3 \cdot d^{-1}$. Flows not included in this expression (between reactors for instance) are assumed to be gravitational with no energy requirement. For all BSMs, the following PE factors are suggested with only the first three applicable to BSM1 and BSM1_LT:

- f_{PE_Qint}: 0.004 $kWh \cdot m^{-3}$
- f_{PE_Qr}: 0.008 $kWh \cdot m^{-3}$
- f_{PE_Qw}: 0.050 $kWh \cdot m^{-3}$
- f_{PE_Qpu}: 0.075 $kWh \cdot m^{-3}$
- f_{PE_Qtu}: 0.060 $kWh \cdot m^{-3}$
- f_{PE_Qdo}: 0.004 $kWh \cdot m^{-3}$

6.3.3 Sludge production for disposal

Sludge production for disposal ($kg \cdot d^{-1}$) is calculated as the mass of solids accumulated in the plant or purposely removed from the process. In BSM1 and BSM1_LT this refers to the sludge removed in the waste activated sludge stream, whereas in BSM2 it is the solids removed from the plant as dewatered sludge. The following equation is used for the calculation:

$$SP = \frac{1}{t_{obs} \cdot 1000} \cdot \left(M_{TSS}(t_{end}) - M_{TSS}(t_{start}) + \int_{t_{start}}^{t_{end}} TSS_x(t) \cdot Q_x(t) \cdot dt \right)$$

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where, $Q_x(t)$ ($m^3 \cdot d^{-1}$) is the sludge flow and TSS_x is the total solids concentration in the sludge flow stream (after dewatering in BSM2). M_{TSS} is defined as the sum of the total suspended solids mass (in kg) present in each of the individual unit processes. For BSM1 and BSM1_LT this includes the activated sludge reactors and the secondary clarifier. For BSM2 the primary clarifier, anaerobic digester and sludge storage tank are also included. M_{TSS} is calculated as follows:

$$M_{TSS}(t) = M_{TSS,as}(t) + M_{TSS,sc}(t) + M_{TSS,pc}(t) + M_{TSS,ad}(t) + M_{TSS,ss}(t)$$

with

$$M_{TSS,x}(t) = TSS_x(t) \cdot V_x$$

and, subscripts 'as', 'sc', 'pc', 'ad' and 'ss' referring to the activated sludge reactors, the secondary clarifier, the primary clarifier, the anaerobic digester and the sludge storage tank, respectively. It should be noted that the total solids concentration in the anaerobic digester (TSS_{ad}) is calculated using the ASM1 states generated in the output stream *after* the ADM1/ASM1 interface. This special case eliminates the need to define a new TSS calculation using ADM1 state variables and clarifies that the calculated TSS throughout the BSM2 layout is based on ASM1 state variables. The reason for this detailed calculation of sludge mass in the systems is to avoid the possibility of temporarily 'storing' sludge in the reactors during the evaluation period, thereby producing less sludge for disposal.

6.3.4 External carbon

External carbon addition (kg $COD \cdot d^{-1}$) is calculated as the total external carbon mass flow:

$$EC = \frac{COD_{EC}}{t_{obs} \cdot 1000} \int_{t_{start}}^{t_{end}} Q_{EC}(t) \cdot dt$$

where, COD_{EC} is the external carbon source concentration ($4 \cdot 10^5$ g $COD \cdot m^{-3}$) and Q_{EC} ($m^3 \cdot d^{-1}$) is the sum of external carbon source flow rates into the biological reactors.

6.3.5 Mixing energy

Mixing energy ($kWh \cdot d^{-1}$) combines energy used for mixing the activated sludge tanks (ME_{as}) with the energy used for mixing the anaerobic digester (ME_{ad}):

$$ME = ME_{as} + ME_{ad}$$

Mixing is assumed to be required in a reactor if that reactor is unaerated or if the applied K_La_i in that reactor is smaller than 20 d^{-1}. Aeration is assumed to provide sufficient mixing when the K_La_i in a reactor is greater than 20 d^{-1}. The energy for activated sludge tank mixing is calculated using:

$$ME_{as} = \frac{24}{t_{obs}} \int_{t_{start}}^{t_{end}} \sum_{i=1}^{i=5} \begin{bmatrix} \text{for } t \text{ when } K_La_i(t) < 20\ d^{-1} \text{ then } f_{ME_{as}} \cdot V_i \\ \text{for } t \text{ when } K_La_i(t) \geq 20\ d^{-1} \text{ then } 0 \end{bmatrix} \cdot dt$$

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where, V_i is the ith tank volume (m^3) and $f_{ME_{as}}$ is the mixing power consumption factor for the activated sludge tanks, which is set to 0.005 $kW \cdot m^{-3}$. It is assumed that the anaerobic digester is mixed constantly, so the mixing energy can be calculated simply as:

$$ME_{ad} = 24 \cdot f_{ME_{ad}} \cdot V_{ad}$$

where, V_{ad} is the liquid volume (m^3) of the anaerobic digester and $f_{ME_{ad}}$ is the mixing power factor for the anaerobic digester which is set to the same value as the activated sludge factor, that is, 0.005 $kW \cdot m^{-3}$. In the BSM1 and BSM1_LT systems ME_{ad} does not exist.

6.3.6 Methane production

Methane production (kg $CH_4 \cdot d^{-1}$) in the anaerobic digester represents an economic benefit and is included in the OCI as a negative value. An average value of the quantity of methane produced per day is derived (based on the general gas law) using:

$$MP = \frac{P_{atm} \cdot 16}{t_{obs} \cdot R \cdot T_{op}} \int_{t_{start}}^{t_{end}} \frac{1}{p_{gas,tot}(t)} \cdot p_{gas,CH4}(t) \cdot Q_{gas}(t) \cdot dt$$

where, $p_{gas,CH4}$ (bar) is the partial pressure of the produced methane gas in the headspace, R is the universal gas law constant ($8.3145 \cdot 10^{-2}$ $bar \cdot m^3 \cdot kmol^{-1} \cdot K^{-1}$), T_{op} is the operational temperature of the anaerobic digester (308.15 K), $p_{gas,tot}$ (bar) is the total gas pressure in the headspace, P_{atm} (bar) is the atmospheric pressure (1.013 bar) and Q_{gas} ($m^3 \cdot d^{-1}$) is the normalised gas flow rate (at P_{atm}) of produced gas. The number 16 represents the atomic weight of methane and t_{obs} is the length of the evaluation period (i.e., 364 days for BSM2).

6.3.7 Heating energy

The heating energy term (Technical Report No. 5) takes into account the energy demand HE ($kWh \cdot d^{-1}$) to heat the anaerobic digester and the heat generated from biogas electricity production. Assuming that 1 kg CH_4 produces 7 kWh of heat, the net heating energy demand can be calculated using:

$$HE^{net} = \max(0,\ HE - 7 \cdot MP)$$

The net heating energy cannot be negative so a correction term is added. It is assumed that the surplus heat that may be produced during electricity generation and that is not used for heating the anaerobic digester is not used elsewhere in the process.

The heating energy HE ($kWh \cdot d^{-1}$) needed to heat the flow of sludge fed to the anaerobic digester is calculated as the average energy input needed to heat the digester inlet sludge flow to the desired temperature in the anaerobic digester:

$$HE = \frac{24}{t_{obs} \cdot 86\,400} \int_{t_{start}}^{t_{end}} \rho_{H2O} \cdot c_{H2O} \cdot (T_{op} - T_{ad,i}(t)) \cdot Q_{ad}(t) \cdot \mathrm{d}t$$

where, ρ_{H2O} is the water density (1 000 $kg \cdot m^{-3}$), c_{H2O} is the specific heat capacity of water (4.186 $kJ \cdot kg^{-1} \cdot °C^{-1}$), T_{op} is the operational temperature of the anaerobic digester (35°C or 308.15 K), $T_{ad,i}$ is the temperature of

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As evaluation experience with this index is still being developed, further tuning of the performance evaluation parameters is needed. However, it should be kept in mind that the selection of these parameters of the index should reflect the benefits and costs associated with the implementation of a fault detection system. Although in the given example only two types of sensor states are considered (normal and faulty) it is possible to extend this index to different types of faults, making this index approach consistent with the benchmark philosophy of providing a platform for investigating new and innovative approaches to simulation problems.

6.6 EVALUATION SUMMARY

An attempt has been made to develop evaluation criteria that are applicable to all of the benchmark models while at the same time providing insight into the evaluation of an implemented control strategy or monitoring methodology. Irrespective of what is being evaluated, the criteria provide a geographically independent means to compare the behaviour of different strategies. Table 6.2 outlines the criteria that are to be calculated for each benchmark model.

Table 6.2 BSM evaluation criteria to be calculated for each model.

Criterion	BSM1	BSM1_LT	BSM2
IQI	✓	✓	✓
EQI	✓	✓	✓
Effluent			
95 percentile	✓	✓	✓
number of violations	✓	✓	✓
violation time	✓	✓	✓
OCI			
aeration energy	✓	✓	✓
pumping energy	✓	✓	✓
sludge production	✓	✓	✓
external carbon	✓	✓	✓
mixing energy	✓	✓	✓
methane production			✓
heating energy			✓
Controller assessment	✓	✓	✓
Risk assessment	✓	✓	✓
Fault detection index		✓	

Although the evaluation criteria are meant to be geographically independent, the structure of the performance measures allows for location specific criteria to be defined in subsequent analyses. That is, the performance measures described above *must* be calculated for each strategy simulation, but emphasis can be placed on specific performance terms depending on location specific criteria if a user so wishes. For example, for a particular user if effluent quality is of primary concern irrespective of overall costs then the analysis of the performance index terms can be weighted accordingly. Alternatively, in another location where reducing overall costs is the primary objective, the index can be tailored to that situation. This structure allows for substantial flexibility in applying the *simulation benchmark* to specific control and monitoring strategies, while at the same time providing a means to make meaningful location specific comparisons and design decisions.

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Chapter 7

Simulation procedure

J. B. Copp, K. V. Gernaey, U. Jeppsson and P. A. Vanrolleghem

The descriptions of the BSM simulation procedures are rigid to ensure the consistent application of the benchmark and to ensure that similar analyses are done on the output data. It is therefore critical that exactly the same simulation procedure be used for each simulated control or monitoring strategy evaluation so that an objective comparison can be made.

7.1 BSM1

For BSM1, a two-step simulation procedure is defined that involves simulations to steady state followed by dynamic simulations using the three influent data files described in Chapter 4.

Steady state simulations

The initial step in the simulation procedure is to simulate the system under study to steady state using an influent of constant flow and composition. The flow-weighted *dry weather* data of Table 7.1 are used for this purpose and steady state is defined using either the software steady state solver *or* by simulating at least 100 days using a constant influent (possibly taking advantage of certain variable step numerical solvers to speed up the calculations). All dynamic simulations should follow a steady state simulation. The steady state simulation should include the same control strategy the user wants to benchmark. However, inactivation of all noise and use of ideal sensor and actuator models should be considered as this will speed up the steady state convergence and will not influence the steady state results. This steady state procedure ensures a consistent starting point and should eliminate the influence of initial conditions on the generated dynamic output. If the investigated control strategy is based on non-stationary control actions (e.g., intermittent aeration) then the non-steady operation should be used during this period to achieve a 'pseudo' steady state before initiating the dynamic simulations.

Dynamic simulations

Next, dynamic simulations should be performed using the influent files described previously. The implementation of these dynamic simulations will vary with the simulator being used, however a general overview of the required procedure is outlined in this section.

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Table 7.1 Steady state influent for BSM1.

Variable	Value	Unit
$Q_{i,stab}$	18 446	$m^3 \cdot d^{-1}$
$Q_{w,stab}$	385	$m^3 \cdot d^{-1}$
$S_{I,stab}$	30	$g\ COD \cdot m^{-3}$
$S_{S,stab}$	69.5	$g\ COD \cdot m^{-3}$
$X_{I,stab}$	51.2	$g\ COD \cdot m^{-3}$
$X_{S,stab}$	202.32	$g\ COD \cdot m^{-3}$
$X_{B,H,stab}$	28.17	$g\ COD \cdot m^{-3}$
$X_{B,A,stab}$	0	$g\ COD \cdot m^{-3}$
$X_{P,stab}$	0	$g\ COD \cdot m^{-3}$
$S_{O,stab}$	0	$g\ O_2 \cdot m^{-3}$
$S_{NO,stab}$	0	$g\ N \cdot m^{-3}$
$S_{NH,stab}$	31.56	$g\ N \cdot m^{-3}$
$S_{ND,stab}$	6.95	$g\ N \cdot m^{-3}$
$X_{ND,stab}$	10.59	$g\ N \cdot m^{-3}$
$S_{ALK,stab}$	7.00	$mol\ HCO_3 \cdot m^{-3}$
T_{stab}	15.00	°C

Starting from the steady (or 'pseudo' steady) state solution, using the *dry weather* influent file as a dynamic input, the system under study should be simulated for 14 days (including noise and non-ideal sensor and actuator models). The resulting state variable values should then be saved (if possible, in the simulator being used) for all unit processes. These state variable values represent the starting point for evaluating the dynamic response of the plant to each of the three influent disturbance files. From the state achieved above, simulate a further 14 days using the *dry weather*, *storm event* and *rain event* influent files in separate simulation studies, but each time starting from the state achieved after the initial 14-day *dry weather* simulation. That is, for any one system at steady state, there are three 28-day dynamic simulations to perform: *dry-dry*, *dry-storm* and *dry-rain*. Where feasible, the user can save all conditions at the end of the first 14-day *dry weather* simulation, reinitiate those conditions in each case and run three 14-day simulations for analysis (*dry, storm, rain*).

The output data generated from the simulations described above are used to examine the dynamic performance of the process. The data of interest from these dynamic simulations are the data generated during the last 7 days of dynamic simulation. That is, in this scenario with 28-day simulations, the data of interest are from the beginning of day 22 until the end of day 28 and include three data sets: one for the *dry weather* simulation, one for the *storm event* simulation and one for the *rain event* simulation. Output data should be recorded at 15-minute intervals for each variable of interest. If the simulation time steps are not exactly aligned with the 15-minute intervals, linear interpolation between simulation time steps should be adopted to calculate the simulation output values at each 15 minute interval exactly.

7.2 BSM1_LT

BSM1_LT was designed as a tool for control strategy or monitoring algorithm evaluation using an evaluation period of one year. The simulation protocol for BSM1_LT is as follows: First, the model is allowed to reach steady state *via* the software steady state solver or by running a 200-day simulation

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with a constant influent. In either case, the simulations should be run without any faults or inhibition. The simulation should be based on the selected control strategy but noise can be deactivated and ideal sensor and actuator models should be used. As the steady state simulation is based on a constant input (Table 7.2) approximating average conditions (using a temperature equal to the BSM1 temperature, 15°C) any time-based control related to temperature effects should be disabled (e.g., seasonal control of the wastage flow rate) and purposefully set to the values the active controllers will have when starting the dynamic simulation. Afterwards, a dynamic simulation is conducted for a period of 609 days using the defined dynamic influent data (flows, concentrations and temperature as defined in Chapter 4). The dynamic simulation is run for 245 days: the first 63 days allows the system to reach a dynamic 'pseudo' steady-state, the next 182 days creates data for training the control or process monitoring algorithms. The first 245 days can be run as many times as desired to tune the algorithms. The final 364 days is used for control and monitoring performance evaluation (days 245–609).

Table 7.2 Influent values for the 200-day stabilisation period when using BSM1_LT.

Variable	Value	Unit
$Q_{i,stab}$	20 850.54	$m^3 \cdot d^{-1}$
$Q_{w,stab}$	300	$m^3 \cdot d^{-1}$
$S_{I,stab}$	27.90	$g\ COD \cdot m^{-3}$
$S_{S,stab}$	54.68	$g\ COD \cdot m^{-3}$
$X_{I,stab}$	48.29	$g\ COD \cdot m^{-3}$
$X_{S,stab}$	190.20	$g\ COD \cdot m^{-3}$
$X_{B,H,stab}$	26.48	$g\ COD \cdot m^{-3}$
$X_{B,A,stab}$	0	$g\ COD \cdot m^{-3}$
$X_{P,stab}$	0	$g\ COD \cdot m^{-3}$
$S_{O,stab}$	0	$g\ O_2 \cdot m^{-3}$
$S_{NO,stab}$	0	$g\ N \cdot m^{-3}$
$S_{NH,stab}$	22.88	$g\ N \cdot m^{-3}$
$S_{ND,stab}$	5.43	$g\ N \cdot m^{-3}$
$X_{ND,stab}$	8.00	$g\ N \cdot m^{-3}$
$S_{ALK,stab}$	7.00	$mol\ HCO_3 \cdot m^{-3}$
T_{stab}	15.00	°C
$S_{tox,stab}$	0	$g \cdot m^{-3}$
$X_{tox,stab}$	0	$g \cdot m^{-3}$

Default control strategies for the reference case are: (1) NO_3-N concentration (class B_0 sensor) maintained in the second compartment at a predetermined set point value (1 g NO_3-N $\cdot$ m^{-3}) by manipulating the internal recycle rate; and, (2) dissolved oxygen concentration (class A sensor) control in the fifth compartment at a predetermined set point value (2 g $O_2 \cdot m^{-3}$) by manipulation of K_La (constrained between 0 and 360 d^{-1}). In addition to these two controllers, carbon, with a concentration of 400 000 g COD $\cdot$ m^{-3}, is added to the first reactor at a fixed rate of 2.0 $m^3 \cdot d^{-1}$. The only other difference with these strategies as compared to the BSM1 definition is the control of the wastage rate. In BSM1, the wastage rate is constant (due to the short evaluation time), in BSM1_LT, two different flow rates are imposed depending on the time of the year (i.e., warm and cold seasons, see Table 7.3).

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Table 7.3 BSM1_LT wastage flow rate as a function of simulation time.

Time (d)	Q_w ($m^3 \cdot d^{-1}$)
$0 \le t < 182$	300
$182 \le t < 364$	400
$364 \le t < 546$	300
$546 \le t \le 609$	400

When the evaluation of process monitoring methods is the goal, sensor and actuator fault models are activated together with inhibitory and toxic material in the influent wastewater. In this case, the predefined input file now contains sensor and actuator fault sequences (obtained from the Markov chain models, Chapter 4) and toxicity events.

7.3 BSM2

The simulation procedure for BSM2 is straightforward. The system is first simulated for at least 200 days using the predefined constant influent data to reach a steady state. The constant influent represents flow-weighted average influent values corresponding to the one year evaluation period; namely, the last 364 days of the 609-day dynamic input data of the BSM2 (Table 7.4). The simulation should be based on the selected control strategy, but noise can be deactivated and ideal sensor and actuator models used. As with BSM_LT, because the steady state simulation is based on a constant input approximating average conditions (temperature equal to 15°C), any time-based control related to temperature effects should be disabled (e.g., season-based control of the wastage flow rate) and purposefully set to the values the active controllers will have when starting the dynamic simulation. If a non-stationary control strategy is used then only a 'pseudo' steady state can be achieved. This procedure ensures a consistent starting point and should eliminate the influence of initial conditions on the generated dynamic output.

Table 7.4 Influent values for the 200 day stabilisation period for BSM2.

Variable	Value	Unit
$Q_{i,stab}$	20 648.36	$m^3 \cdot d^{-1}$
$Q_{w,stab}$	300	$m^3 \cdot d^{-1}$
$S_{I,stab}$	27.23	$g\ COD \cdot m^{-3}$
$S_{S,stab}$	58.18	$g\ COD \cdot m^{-3}$
$X_{I,stab}$	92.50	$g\ COD \cdot m^{-3}$
$X_{S,stab}$	363.94	$g\ COD \cdot m^{-3}$
$X_{B,H,stab}$	50.68	$g\ COD \cdot m^{-3}$
$X_{B,A,stab}$	0	$g\ COD \cdot m^{-3}$
$X_{P,stab}$	0	$g\ COD \cdot m^{-3}$
$S_{O,stab}$	0	$g\ O_2 \cdot m^{-3}$
$S_{NO,stab}$	0	$g\ N \cdot m^{-3}$
$S_{NH,stab}$	23.86	$g\ N \cdot m^{-3}$

(*Continued*)

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Table 7.4 Influent values for the 200 day stabilisation period for BSM2 (*Continued*).

Variable	Value	Unit
$S_{ND,stab}$	5.65	$g\ N \cdot m^{-3}$
$X_{ND,stab}$	16.13	$g\ N \cdot m^{-3}$
$S_{ALK,stab}$	7.00	$mol\ HCO_3 \cdot m^{-3}$
T_{stab}	14.86	°C

The steady state values obtained in this first simulation are subsequently used as initial values for simulations using the dynamic influent and the wastage flow rates of Table 7.5. From this starting point, BSM2 is simulated for 63 days using the dynamic influent with controls active (including noise and non-ideal sensor and actuator models) to achieve a 'pseudo' steady state. This period is followed by 182 days of dynamic simulation. This 'training' period is used to tune adaptive or model-based controllers, to set internal parameters or to tune the control algorithms in another way. Once training is complete, BSM2 is simulated for an additional 364 days with the output data (stored at 15-minute intervals) used for plant performance evaluation. This final period is only simulated once the algorithms are suitably tuned/trained and need not be simulated while testing. Each control strategy, regardless of the study objective, is simulated in precisely the same way to ensure standardised results.

Table 7.5 BSM2 wastage flow rate as a function of simulation time.

Time (d)	Q_w ($m^3 \cdot d^{-1}$)
$0 \le t < 182$	300
$182 \le t < 364$	450
$364 \le t < 546$	300
$546 \le t \le 609$	450

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Chapter 9

BSM limitations

J. B. Copp, K. V. Gernaey, U. Jeppsson and P. A. Vanrolleghem

The use of the benchmark systems for the assessment of process performance, control system evaluation and similar purposes is well established and the repeated reference to the first COST/IWA benchmark simulation model (BSM1) has clearly indicated the usefulness of such a tool for this purpose. However, the benchmark effort is not without its critics. The unit process sizes and model choices, influent characterisations, model transformations and evaluation criteria have all been criticised. Some of the criticism is justified as the defined BSMs have not always taken into account the most recent advancements or accepted theory, but the critics often fail to fully appreciate the benefits generated by the effort, the debates, the compromises and the solutions that have gotten the BSMs to this point. The BSM development group has ebbed and flowed over the years, but the group was typically made up of 20 or more modelling experts from around the world discussing the BSM development at regular meetings. What is further impressive is that this development has been carried out on a voluntary basis. The group, however, does recognise that the BSMs have limitations and even though this is consistent with any other mathematical models, it is important that benchmark users understand those limitations in order to assess the validity of their simulation studies. This chapter addresses those limitations.

9.1 BSM AS A TOOLBOX

About two decades ago, the concept of a tool that could be used to objectively evaluate the performance of control strategies through simulation using a standard model implementation was introduced for activated sludge wastewater treatment plants. That concept resulted in the development of BSM1, the subsequent BSM1_LT and most recently BSM2. Practitioners have suggested that these models are only academically applicable, have been conceived of for publication generation purposes and provide limited benefit to the applied modelling community. The authors of this STR beg to differ with those practitioners. It has further been suggested and it is recognised that control strategies that work well in a BSM may not work as well in practice due to the many factors affecting a full-scale plant. As authors of this STR, we are fully aware of this. Here, again it is important to keep in mind that the main purpose of the BSM development has been to create a tool for objective evaluation of control strategies through simulation. It has never been the primary plan to solve problems on a real plant with the BSMs. In addition to the successful use of BSMs

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for objective control strategy evaluation, the development of the BSM systems has provided a number of spin-off benefits and has resulted in the development of a number of universally applicable sub-models. The BSMs can be used in their entirety, but to limit their application this way would hide the value of the pieces that make up these models. Rather, the BSM systems also should be considered a modelling toolbox, and a platform on which modelling issues have been debated, experimented upon and tested.

The BSM sub-models should be considered a collection of modelling tools that address various aspects of whole-plant wastewater treatment modelling. The BSMs contain a whole series of these modelling tools including but not limited to: an influent wastewater generating model, a temperature model, empirical solids/liquid separation models, standard and verified implementations of ASM1, ADM1, the Takács double exponential settling model and the Otterpohl primary clarification model, anaerobic digestion/activated sludge model interfaces, a risk assessment model to infer solids separation problems of microbiological origin, performance indices, operational cost indices as well as models for energy consumption by aeration and pumping equipment and generic models for sensors and actuators. The BSM development value lies in these individual modelling tools, and the modular nature of those tools means that they are portable and can be used in isolation if the need arises. These tools have all appeared as a direct result of the BSM development, and without that effort, may not have appeared at all. Not only that, these tools have formed the basis for the development of standardised approaches to modelling issues like model interfaces and simulation speed optimisation as published beyond the BSM context (Nopens *et al*. 2009; Rosen *et al*. 2006; Volcke *et al*. 2006b). Some of the tools, (i.e., the ADM1 implementation (Technical Report No. 5) and the influent wastewater generation model (Gernaey *et al*. 2011)) are available as separate tools upon request, but all the tools resulting from the development are distributed for free and many come as standard components of the commercial WWTP simulators.

Model verification through ring-testing identified a number of implementation issues, coding errors and simulator specific insights, and it is quite likely that many of the identified issues would have remained unresolved in the commercial simulators had this effort not taken place. The ring-testing was instrumental in identifying minor problems under normal plant operating conditions. Extreme conditions were not tested, so it is unclear if other numerical issues still exist. However, one thing that was clearly illustrated through this process was the need to ring-test all new implementations. The Task Group was able to test the predominant software packages currently on the market, but undoubtedly new software packages will come to market in the future and users should ensure the quality of those software packages by testing them with the BSMs. Although it is comforting to assume that the software developers have implemented error-free code, the reality is that error-free code is rare, so caution is appropriate and the verification approach developed and applied in the BSM-work can help in minimising such errors in the future.

Users should also be cautioned that certain model implementations were heavily scrutinised and others were not tested as rigorously. The ring-testing effort was initiated to test the biokinetic model implementations and was later expanded to include the other unit process models in each software platform. The biokinetic models were extensively tested, but other models like the influent generator, or the calculation of the evaluation criteria were not evaluated to the same extent. As a result, users are cautioned to check the implementation of all sub-models in their platform of choice to ensure that simulator-specific issues will not affect the calculated results.

9.2 MODEL STRUCTURES

9.2.1 Biokinetic models

The BSMs are not all-encompassing tools nor are they 'best-practice' tools to be interpreted as showcasing the best models for specific unit processes. Many of the published models used in the BSMs were chosen

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based on compatibility and international acceptability at the time. The Task Group is not advocating the use of these models for purposes unrelated to benchmark studies as it is fully recognised that some of the benchmark models have specific short-comings.

For example, since its first introduction, ASM1 has been criticised because the published model does not contain ammonia switches in biomass growth rate expressions. Equally unfortunate is the omission of inorganic solids in the original description. Similarly, there is evidence suggesting that pH can affect the activated sludge activity, but pH is not considered in the published ASM1. The same holds true for processes such as enhanced biological phosphorus removal (EBPR), 2-step nitrification and denitrification including N_2O production and Anammox extensions to ASM1. Although these extensions and oversights have been included in many commercial simulator implementations, the benchmark description of ASM1 is the original published version.

The anaerobic digester in BSM2 is modelled using ADM1 (Batstone *et al.* 2002). However, it was necessary to implement a modified version of the published ADM1 to optimise the simulation performance and correct some mass balance issues (Rosen *et al.* 2006). Several changes were implemented including continuous inhibition functions for pH and an algebraic solution for the hydrogen state variable. The changes were the direct result of the BSM development and reflect the extensive investigation of ADM1 as it related to the simulation of the benchmark model. These changes represent significant improvements to the published model for the BSM2 implementation specifically and for practical simulation applications in general. Readers may want to adopt these changes for other non-BSM studies, but should do so while recognising that the changes were developed for BSM optimisation and may or may not be appropriate for every application of this model.

The inclusion (compared to BSM1) of primary treatment as well as sludge treatment in BSM2 increases the complexity of the system but more importantly allows for the study of unit process interactions. These inclusions have forced the benchmark team to develop and consider a new set of modelling tools that have far reaching implications and uses. Because the activated sludge and anaerobic digestion models each use different state variable sets, AS/AD interfaces were developed to account for a number of aspects including COD and nitrogen mass balances, charge continuity and differences in digester performance depending on the primary and secondary sludge composition of the digester influent. Here again, the interfaces were developed based on *a posteriori* knowledge of the system, the digester influent and the performance of the various unit processes. That is, a great deal of debate surrounding these interfaces was directed at the partitioning of the COD into appropriate state variables. A generalist approach was initially adopted, but this approach quickly identified performance issues that were deemed unrealistic in the context of BSM2. Based on this, the approach was modified with the goal of obtaining a particular result and the adopted model achieved that goal. As a result, the interface models may not be ideal for other applications. As with the other models, the Task Group believes that the adopted approach is applicable and that the model parameters are reasonable, but model users should realise that these parameters may not be applicable for all situations.

9.2.2 Aeration

Since first implemented into software tools, aeration models have been criticised for the simplifications that have been adopted. Many of the commercial simulators have updated their implementations and now include efficiency calculations in their transfer rates. That is, some of the more complicated models now attempt to predict variable transfer efficiencies based on the airflow and the airflow/diffuser. This is not the case for the BSM models. In the BSMs, aeration is modelled by directly varying the K_La. That is, if a DO controller calls for more oxygen, the model responds by increasing the K_La. The K_La is then

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translated into an estimated aeration energy quantity which is the output from the model. In reality the relationship between K_La, airflow and transfer efficiency is not that simple and experience tells us that blowers operating in an inefficient range can add significant costs to plant operations. So, users may need to consider this limitation if aeration control is a significant result from the control strategy that is being evaluated.

9.2.3 Solid/Liquid separation models

Several limitations related to the solid/liquid separation models are acknowledged. The thickening and dewatering units in BSM2 are assumed to be ideal. This is clearly not the case in reality but this assumption was deemed sufficiently accurate for BSM2 purposes. The Takács double exponential settling velocity model in a layer model is recognised as a reasonable representation of the settling process, but its inability to predict the impact of short term settling issues and compression makes this model unsuitable for judging absolute settling behaviour. That is, this model is ideally suited for examining relative changes in the predicted settling and sludge blanket behaviour and can be used to judge the impact that might result at a real facility, but the absolute changes in effluent solids predicted by the model may not be accurate. Furthermore, the adopted settling models do not include any biological conversions which may or may not be significant. The exclusion of biological activity in the clarifiers is a simplification that was deemed reasonable given the low sludge blankets in the BSMs, but different operating conditions could result in different settling conditions and it is widely recognised that denitrification in clarifiers can be significant, especially if a large blanket exists (e.g., Siegrist *et al.* 1995). Users should consider these limitations when performing benchmark simulations.

Settling problems can also be caused by a number of different things including hydraulics and microbial population shifts. These issues are not typically modelled and they are not included in the Takács model. However, it is widely recognised that bulking, foaming and rising sludge can be significant problems in reality. To address this issue, the BSMs have included a new risk analysis formulation that attempts to mathematically predict the propensity of a given treatment scheme to have settling problems (see section 4.5). The development of this approach is based on both the available literature and empirical knowledge used in practice, but the validity of the approach still needs to be evaluated with full-scale data. That is, issues like weir design that contribute to filamentous proliferation are not included in the model so it is quite possible that each model application will be unique.

9.2.4 Other models

Limitations exist in the other models as well. Pumps, for instance, are assumed to be ideal in the basic BSM implementations. That is, if a pump set point is changed, the pumped flow is assumed to change instantaneously without error to the new rate unless an actuator model is used to control the change. This simplification has the potential to generate unrealistic output as real pumps do not operate in this way, nor do they pump precisely constant flows over extended periods of time. Care should be taken by the BSM user if the control strategy under study relies heavily on a pumped flow control scenario. The sensor and actuator models that were developed as part of the BSM1_LT work allow including this more realistic behaviour, dynamic response to changes, delays and drift so as to mimic reality better. They come at a cost, however, as more implementation work is required and they typically result in slower computations.

The modelling of sensors is slightly more detailed as signal response times are modelled. It was reasoned that if a control strategy required a sensor and the sensor signal delay was significant, then the evaluation should take this into account. For example, the evaluation of a respirometry-based control strategy would

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certainly be unrealistic if the OUR measurement on which it was based was assumed to be instantaneous when in reality the actual measurement might take 10, 20, 30 minutes or longer to acquire. However, as the BSMs were developed so that control strategies could be compared, it was necessary to limit the permutations that a user could choose when modelling a sensor. Much like the reasoning behind choosing the biological parameters, it was not appropriate (as the results would have been too hard to compare) to allow users the option to choose all of the parameters for a particular sensor without limiting those choices in some way. In this case, sensor classes with specific features were defined to limit those choices (Rieger *et al.* 2003). This is, therefore, by definition a limitation and the user should recognise that it is more than possible that these classes will not be explicitly consistent with a particular sensor. As a result, users will be forced to choose a sensor class that has characteristics that are the closest match to their sensor. This limitation may skew the results slightly, but by limiting the sensor definitions, the results of any particular BSM study will be more readily comparable to other studies.

Finally, the way the temperature dynamics in the bioreactors is modelled in BSM1_LT and BSM2 is kept very simple. The main temperature dynamics in the influent follow a sinusoidal variation over the year, which may or may not be accurate and the temperature in combined streams is based on a flow-weighting equation. No heat generation or losses are accounted for. Furthermore, BSM2 assumes a constant temperature in the anaerobic digester which may or may not be significant for some strategies.

9.3 MODEL PARAMETERS

The model parameters defined in the BSMs are for control evaluation purposes only. The chosen parameters are believed to be reasonable for all of the unit processes, but the Task Group is not suggesting that these parameters be used for any other modelling purpose. That is, each of the sub-models has a number of defined parameters. These parameters have been chosen for the BSMs to achieve certain goals like removal efficiencies or are based on certain assumptions like temperature (e.g., 15°C for BSM1). The Task Group makes no claim as to the validity of those parameter value choices, nor does the Task Group purport that the assumptions on which those parameters are based are correct. The Task Group believes those assumptions to be reasonable, but the choice of model parameters outside the benchmark environment should be based on sound, local knowledge related to the modelling task at hand.

For example, BSM1 assumes a temperature of 15°C and the defined parameter values in that system reflect parameter estimates at that temperature. BSM1_LT and BSM2 have a variable temperature so temperature corrections based on a temperature correction equation have been adopted for the kinetic parameters. These corrections have been assumed on the basis of literature information so they may not be the best estimates for those parameters at any given temperature. The digester is assumed to have a constant temperature, but the activated sludge portion of the model has a variable temperature, mainly driven by the influent temperature. The Task Group believes that these are reasonable assumptions and simplifications, but the reader should not deem these as the best or only approach for other applications. Any temperature modelling done outside of the BSMs should consider local and appropriate assumptions where necessary, but the approach adopted in the BSM development may provide a reasonable place to start in the absence of more detailed local information.

9.4 EVALUATION CRITERIA

The development of the evaluation criteria required a significant amount of discussion, debate and compromise. Through this process, the Task Group and various contributing research groups were enlightened as to the many different effluent regulations that exist across the globe. Some groups advocated for mass load

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limits, while others wanted every gram of effluent pollution counted. Some groups wanted 'soft' limits like 95 percentiles and others were content with firm concentration limits. As the goal was to develop criteria that were location independent, it was agreed that the use of concentration limits was the best approach. However, here again, it was not straightforward as to what those concentration limits should be. In the end, the limits were arrived at through a debate and compromise and are not meant to reflect limits in a particular location.

Similar discussions surrounded the other criteria as well. The EQI factors are loosely based on Flemish legislation (Vanrolleghem *et al.* 1996) and the energy calculations are the result of various proposals. For example, because the options for real pumping equipment are considerable and the characteristics of that equipment vary, it was necessary to make a number of assumptions to achieve a pumping energy number for the OCI calculation. This process was repeated for the aeration, mixing and heating energy calculations. As a result, it is important to note that the calculated quantities may not be completely accurate, but instead should be considered to be relative terms. That is, if the OCI increases from one scenario to another, it is presumed that the operating costs will rise if those changes were made at full scale. The absolute increase in operating costs may not be known, but the relative comparison of the costs suggests that an increase is likely. A similar conclusion can be reached about changes in the calculated EQI. An increase in the EQI suggests that, relatively speaking, the effluent quality will deteriorate.

A significant amount of debate during development surrounded capital cost evaluations. The implementation of control is expected to require a capital cost allocation. Some plants will already have the infrastructure in place, but most plants will not. As capital costs have not been included, it is possible that a control strategy with an unrealistic capital expenditure could be developed. This strategy could result in an evaluation criteria output that is exceptional without being penalised for the capital costs. This is a limitation that users should be aware of. In reality, the gain in OCI of the control strategy might be compared to the capital expenditures, for instance in terms of a payback period or an internal rate of return, but this would depend on the local situation. Indeed, the major problem with estimating capital costs is that they are often location specific and a conscious effort has been made to avoid location specific criteria. This results in a conflict that has been addressed qualitatively rather than quantitatively. When evaluating a control strategy, it is recommended that users include a list of new equipment (sensors, actuators, pumps, tanks, pipes,...) that is required for the strategy. This list may not be comparable to the other evaluation criteria directly, but it will make the comparison of a strategy that reports a modest operational cost index gain with an extensive equipment requirement list comparable to another strategy with limited infrastructure requirements. It is important always to keep in mind that the evaluation criteria represent relative changes and are simply indicative of potential differences between one strategy and another and this list achieves that goal.

9.5 MODEL SIMULATION

The evaluation criteria in all of the BSMs are based on an arbitrary data resolution of 15 minutes. This frequency provides sufficient resolution to ensure that the simulation dynamics will not be missed. It was reasoned that if the output was recorded less frequently, underlying plant dynamics could occur in between sampling times in which case the simulation output could miss an event completely. The opposite is also true as less frequent output could result in peak measurements only and miss the valleys. More frequent output (say at 1 minute intervals) was considered as well, but ultimately this was rejected because the added data was thought to provide little if any benefit to the dynamic evaluation. Using a 15 minutes data resolution was deemed to be a reasonable compromise between the amount of data output, simulation speed and event resolution.

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BSM1 was developed using a 7-day evaluation period. From a control perspective this may well be sufficient if the control loop is fast and the impact of the control action is equally rapid. That is, aeration and recycle pumping control loops have sufficiently short time constants to be analysed over a 7-day period. When a dissolved oxygen control loop calls for more air, the blowers ramp up and more air is supplied within minutes. If a recycle pump is turned up, the impact on the process can be seen almost immediately in the simulation output, even though the full impact on the sludge concentrations follows the hydraulic retention time. However, three hydraulic retention times fall still within the 7-day period. So, for fast control loops like these, 7 days may well be sufficient.

However, a limitation with this time period presents itself when control loops with longer time constants are evaluated. For example, control strategies that aim to control the solids retention time will need far more time than 7 days of evaluation because the impact of changing the waste sludge flow rate may take more than 10 days to materialise in the simulation output. Seven days is equally problematic for evaluating stochastic events like sensor or actuator faults. Different types of faults will have different and time varying impacts and 7 days may not be sufficient to observe those effects. Modelling temperature impacts cannot be done in a 7 day period either. The liquid temperature in a treatment plant varies gradually over time and may be at its highest or lowest for only a short period of time over the course of the year. So, it is unrealistic to model larger temperature changes over that short a period and it may be equally unrealistic to model a low temperature steady state solution as a real plant never actually experiences that temperature long enough to reach steady state. To address this issue, BSM1_LT and BSM2 have adopted a one year (52 weeks, 364 days) evaluation period. Using a period of this length provides the BSM user with the option to evaluate more control strategy options that could not be effectively evaluated using 7 days and observe the effect of seasonality in temperature and precipitation. Regardless of the evaluation time being used, the BSM user should be aware of the limitations introduced by the length of these evaluation periods.

9.6 APPLICATION EXTENSION

The use of the modelling toolbox provides an excellent starting point for modelling and evaluating many systems. Examples of such applications have been presented by Abusam (2001), Volcke *et al.* (2006a) and Benedetti *et al.* (2006). The ability to 'extend' the BSMs is an unquantifiable benefit of the work and shows that the BSM influence should not be restricted to control strategy evaluation alone. Rather, this work has illustrated its importance and associated benefits to modelling in general.

The toolbox has been freely distributed to research groups on all continents to provide a structured, documented and validated starting point for their future work. The development of the BSM systems has also provided a number of spin-off benefits beyond the models themselves. This development effort has been criticised in some circles as having limited value due to the theoretical approach and the fact that the model is not based on any particular plant. However, these models were developed as a means to develop and test control strategies in an unbiased fashion. That is, although there are numerous literature studies on the effectiveness of various control strategies, it is typically not possible to compare these studies because the conditions used to test the strategies are different. To address this issue, the descriptions of the BSM simulation procedures are rigid to ensure the consistent application of the benchmark and to ensure that similar analyses are done on the output data for each simulated control strategy. Only with a completely defined simulation procedure can any conclusions be reached when one strategy is compared to another. So, although some of the models already existed, the protocol for the standardisation of the output evaluation is an interesting development. The evaluation criteria are not exhaustive, incorporate a number of limitations and have many inherent assumptions, but they do represent a standardised approach.

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The BSM definitions discussed in this STR are rigidly based on a pre-denitrifying activated sludge system in a Modified Ludzack-Ettinger (MLE) configuration (this is the default set-up of the BSMs although available control handles allow for post-denitrification systems, various step-feed systems, anox-aerobic-anox-aerobic systems and many others). Because of this rigidity in the definition, the application of the BSMs, that is, the evaluation of control strategies, is somewhat limited to plants with this configuration, which is only a subset of the plants in operation. During the initial stages of the BSM development, Copp *et al*. (2002) expanded the COST benchmark work (Copp, 2002) to include nitrifying and carbon removal only configurations. Although the models developed in that work are not entirely consistent with the BSM described here, it does show that portions of the BSMs can be modified for specific applications if the need arises. Extensions have also been made within the BSM framework to evaluate oxidation ditches (Abusam, 2001; Abusam *et al*. 2002), Sharon (Volcke, 2006), Anammox (Volcke *et al*. 2006a) and MBR systems (Maere *et al*. 2011). These BSMs are fully defined, but as the application of the framework expands, it is fully expected that new extensions and configurations will be developed for other processes such as EBPR (Gernaey & Jørgensen, 2004), simultaneous nitrification/denitrification and greenhouse gas emissions (Flores-Alsina *et al*. 2011; Guo & Vanrolleghem, 2014) and the fate of micro-pollutants (Snip *et al*. 2014). Although, defining the configuration is a limitation, the modular nature of the framework allows for future development and should enable continued use of the modelling toolbox.

9.7 CONCLUSION

The BSM development has been an international collaborative effort that has resulted in a number of new model initiatives. This effort has provided significant insights into the modelling of wastewater treatment from a mathematical and numerical point of view. It has set the scene for proper implementation procedures and ring-testing schemes. It has produced a number of new models (for influent, sensors, error, interfacing and evaluation) or important modifications (ADM1) and has provided standardised simulation protocols. It is critically important to realise that none of these models have been calibrated to real data from a full-scale facility (within the framework of the BSM development) and that the processes being modelled are not based on a real plant. However, it is equally important to point out that the BSMs represent years of discussions, debates and compromises by a comprehensive, global expert team and although not perfect they have identified and focused attention on numerous modelling issues that did not have standardised solutions prior to this work.

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Chapter 10

Conclusions and perspectives

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This Scientific and Technical Report (STR) summarises the result of approximately 20 years of development work on the different benchmark platforms. Apart from the benchmark platforms themselves, this Task Group has been able to produce a set of verified and platform independent unit process models that are essential for WWTP control strategy evaluation simulation studies. Furthermore, the Task Group has spearheaded the development of a set of generic simulation tools that can be used to support the modelling of activated sludge WWTPs in general. In this chapter, conclusions are drawn on the basis of our experience with the development and the usage of the benchmark platforms and an outlook towards future developments is provided.

10.1 LESSONS LEARNED: DEVELOPMENT OF THE BENCHMARK PLATFORMS

As discussed throughout this STR, the benchmark platforms – BSM1, BSM1_LT and BSM2 – are comprehensive tools and freely available simulation protocols which allow for the objective comparison of WWTP control and/or monitoring strategies.

It is important to emphasise that the development of the benchmark platforms was a voluntary effort for the most part. Considering the large number of scientific papers that have appeared on the development and the use of the benchmark platforms, one important lesson learned is that voluntary work still has a major role to play in the scientific community. Of course, there has been some financial support for meetings and the Task Group is grateful for the support from IWA and the COST actions 682 and 624. Clearly, the development of the benchmark platforms has demonstrated that such support (e.g., COST actions) is important for promoting frequent contact between researchers working in similar areas. This development would not have been possible without that support. Regular face-to-face discussions are indeed essential when embarking on this type of journey and these meetings also facilitated the interaction of MSc and PhD students whose projects were modulated to fit into the benchmark development project.

Another lesson learned from this development is that collaboration between industry and academia is essential for a successful outcome. Indeed, the fact that the development has been supported by a broad and diverse group of people is key to its success in the wastewater modelling and simulation

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community, and this has been assisted by the fact that the development, from the outset, has been simulation platform independent.

The benchmark development also has resulted in many spin-off benefits. The first indication of these benefits was the ASM1 ring-testing – the confirmation of results using different software platforms and different developers – that was carried out very early on in the BSM1 development. Obscure errors were discovered and simulator-specific issues were identified. Without the rigorous testing that BSM1 required, it is not unreasonable to conclude that those errors might still exist in the commercially available software packages. Ring-testing of all the unit process models has, in general, been a very time-consuming exercise, and has been one of the most challenging tasks, but it has provided some of the most important spin-off benefits. Further benefits became apparent during the BSM2 development because a significant amount of time was spent on model development issues such as the ADM1 implementation, an influent wastewater generator model, ASM1 → ADM1 and ADM1 → ASM1 model interfaces, in addition to several others. The effort put into these issues is important because they are generic tools that might be (and already are) used outside of the benchmark environment.

The expansion of BSM1 to include primary clarification and sludge treatment with anaerobic digestion for BSM2 resulted in some interesting developments from a control point of view. An initial draft of BSM2 with the activated sludge tank volumes identical to the BSM1 volumes was presented at WaterMatex2007 (Jeppsson *et al.* 2007). At that time, it was shown using 15 simple control strategies that control of that plant was limited and control authority was insufficient to significantly improve the performance of the highly overloaded plant layout. Interestingly, the high nitrogen load that was causing some of the issues was associated with the reject water, which was not present or accounted for in the BSM1 case. The activated sludge tanks in BSM2 were thus redesigned and the current layout was presented several years later by Nopens *et al.* (2010). This layout (compared to the earlier versions) included: (1) a reduced influent nitrogen load to compensate for the reject water contribution; (2) increased activated sludge tank volumes, compared to BSM1, in order to obtain a WWTP that can benefit from process control; and (3) modifications to the evaluation criteria to distinguish between nitrate and ammonia nitrogen in the effluent quality index, to reduce the aeration impact on the calculated cost index and to differentiate the energy consumption for different pumped flows.

10.2 LESSONS LEARNED: USE OF THE BENCHMARK PLATFORMS, VERIFIED PROCESS MODELS AND GENERIC TOOLS

The benchmark platforms have been well received by the research community. BSM1 and BSM2 are used frequently for their original purpose (e.g., Vrecko *et al.* 2002; Stare *et al.* 2007). They are relatively easy to use because the different parts (unit process models, control strategies, evaluation criteria,…) are well documented and tend to be immediately recognisable by most users. The concept of benchmarking monitoring strategies, as introduced in BSM1_LT, has been used less frequently thus far, but this benchmark provides a necessary platform nevertheless in a niche area of research. The Task Group hopes that BSM1_LT will receive more interest and be a source of inspiration in the future, as monitoring strategies and methods become increasingly important for WWTPs.

10.2.1 Portability

The modular construction of the BSM platform is one of the main reasons for its success. Because the benchmark can be viewed as being the combination of several generic tools, it is relatively easy for a user to pick one or more of these tools and use them in isolation or as part of another study. Furthermore, this

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structure gives users the ability to add on one or more additional features to the BSM platform without the need to implement a significant amount of computer code. This is illustrated with some examples below.

Quite recently, Sin *et al.* (2009, 2011) used the BSM1 plant layout to demonstrate the application of sensitivity and uncertainty analysis methods. This application was not envisioned when the benchmark development was initially discussed, but is an interesting illustration of how the BSM platform has been used as a convenient vehicle for the demonstration of new ideas or tools.

In other cases, benchmark users have not used the BSM layouts, but have instead used one of the benchmark tools (e.g., the evaluation criteria) to compare control strategy performance on models of other plants. Abusam *et al.* (2002) was one of the first to apply the BSM1 evaluation criteria to another plant model. Many similar applications of the evaluation criteria can be found in the literature. Other authors have used the BSM1 influent data files (dry, storm and rain weather data) or transformations thereof in order to simulate a reasonable dynamic influent.

The Task Group is pleased that users take parts of the BSM platform, and use these in their research projects. However, it is critical that users clearly document what parts they have used and what transformations they have implemented. Unfortunately this is not always the case and the documentation in some studies is insufficient. It is important that the users specify precisely which parts of the BSM platform have been used and also provide a detailed description of any additions. This is essential to ensure that other users understand what has been done and can duplicate the work if necessary.

10.2.2 Extensions

With the extension of BSM1 to the plant-wide BSM2, the number of potential extensions has increased substantially. Volcke *et al.* (2006a) were the first to modify the BSM2 layout with reject water treatment. More recently, the Task Group has observed quite a number of publications where the BSM2 plant layout was extended with extra functionality. For example, one known limitation of the BSM biological model is that only one-step nitrification and denitrification are used, but there has been a growing interest in predicting greenhouse gas formation from the treatment process, and N_2O – an important nitrification and denitrification intermediate with a high greenhouse gas (GHG) potential – is one of the compounds that is being studied extensively. Multi-step nitrification/denitrification models including N_2O production have already been implemented in the BSM2 benchmark framework (Flores-Alsina *et al.* 2011; Corominas *et al.* 2012; Guo *et al.* 2012; Guo & Vanrolleghem, 2014). Several other potential extensions have been described by Jeppsson *et al.* (2013). In many cases these add-ons have been defined, implemented and remain the property of individual research groups which contrasts the Task Group's philosophy that verified implementations of the benchmark plants are to be freely distributed. Clearly, the Task Group supports such extensions. It is certainly hoped that these extensions will be made available in the future, and that the documentation of such extensions will be extensive so that the entire research community can benefit. The Task Group invites all interested parties to develop additional tools, but emphasises that model extensions should be ring-tested (e.g., by comparison of two or three independent implementations by different users) before being distributed. Ring-testing these extensions is the best way to ensure the quality of the documentation and the computer code. Although not currently ready, the Task Group is discussing the definition of a suitable protocol to ensure the quality of any additions.

10.3 LOOKING AHEAD: FUTURE EXTENSIONS OF THE BSM PLATFORMS

One major conclusion is that the benchmarking work is far from finished. A wish list for future extensions to the BSM platforms was presented in June 2011 at the Watermatex conference in San Sebastian (Spain)

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by the Task Group chairman, and this work was later published in a journal paper (Jeppsson *et al.* 2013). Potential extensions can be subdivided into several categories including: (1) Temporal extensions; (2) Spatial extensions; (3) Process extensions within the WWTP; (4) Realism of the models used in the BSM; (5) Control strategy extensions; and (6) Extended evaluation tools. The most important trends are briefly highlighted here.

10.3.1 Temporal extensions

Dynamic simulations with BSM1 were based on 14-day influent files depicting dry, rain and storm conditions, but this approach was shown to be insufficient for long-term control applications such as sludge age effects and equipment failure scenarios. To address this shortcoming, a phenomenological influent wastewater generator model was developed to cope with the challenge of providing realistic long-term (1–2 years) influent data to the BSMs. Even longer influent data files might be required for future scenario analyses. For example, the simulation of changes in urban catchments and the impact on treatment plant efficiency might require evaluation periods in excess of ten years. The ever-increasing computational power would allow an extended evaluation period; however, the practical relevance of results based on such extended simulations would likely be highly uncertain. Nevertheless, such an extension may be well-received in the future.

10.3.2 Spatial extensions

The benchmark platforms described in this STR are defined as 'within-the-fence' systems, that is, the model descriptions and simulations do not extend outside the borders of the WWTP. The importance of the sewer system and processes in the receiving waters were recognised by the Task Group but including these complicating factors in the original BSMs was deemed to be beyond the Task Group's scope. From a control and monitoring perspective the inclusion of the sewer network into the benchmark system would open up a range of new possibilities for interactions and manipulation of the combined sewer/WWTP system (e.g., back-flow effects, storm tanks and pumping stations, combined sewer overflows, pollution contributions from run-off). Examples of integrated sewer and WWTP modelling can be found in Sharma *et al.* (2012) and an initial combined benchmarking study connecting the sewer model with BSM2 was performed by Guo *et al.* (2012). As for receiving waters, existing models such as the River Water Quality Model No. 1 (Reichert *et al.* 2001) could be added or linked to the BSMs without too much difficulty given that the model interfaces have already been developed (Benedetti *et al.* 2007). This kind of approach would be particularly beneficial for a more detailed evaluation of the environmental impact of wastewater pollutants. Moreover, such a combination would promote the use of the benchmark system as a decision support tool in agreement with current river basin management approaches, as pursued for example by the EU Water Framework Directive.

Integrated evaluation experience can be found in Benedetti *et al.* (2010) and Brehmer *et al.* (2009). As with the benchmarking tools developed so far, consensus will have to be reached on objective evaluation criteria that assess the urban water quality impacts in receiving waters, but ideas for this are being developed (e.g., Bauwens *et al.* 1996; Vanrolleghem *et al.* 2005a; Benedetti *et al.* 2010; Clouzot *et al.* 2013).

10.3.3 Process extensions

Process extensions within the WWTP are related to the appearance of models for new unit processes, such as the SHARON and ANaerobic AMMonium OXidation (Anammox) processes (Volcke *et al.* 2006a). The Task Group expects that this evolution will continue in the future. The Task Group has recognised an

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immediate need for process extensions related to integrated fixed-film processes (Vanhooren *et al.* 2002) and SBR configurations. Note that these process extensions will have to be coupled to suitable model interfaces if the state variables in the new model differ from the state variables in the original benchmark models. As discussed previously in this STR, these interfaces ensure that material mass balances and continuity principles are met when mapping the output variables of one model to the most appropriate input variables of another model (Alex *et al.* 2005; Vanrolleghem *et al.* 2005b; Volcke *et al.* 2006b; Nopens *et al.* 2009) and these issues are critical for acceptance within the modelling community.

10.3.4 Realism of models used in BSM

The mathematical models used in the benchmark platforms were chosen because they are internationally accepted and well established. The Task Group is aware of the limitations in the chosen models; but, is also keenly aware of the fact that there is an almost limitless possibility to extend and upgrade the models within the existing BSM plant configurations, including the models describing sensors and actuators (e.g., more detailed and dynamic models of blowers and pumps). It is unequivocally hoped that the aim of any changes would be to enhance realism of the systems rather than to simply increase the level of detail and complexity. In some cases, improved models have become available and although the Task Group decided not to change models during the development phase, most of these updated models are well described, and could be interchanged within the current BSM framework. For example, future extensions may include phosphorus removal, multi-step nitrification/denitrification and N_2O production, inclusion of sulphur reducing/oxidising reactions, improved thickener and dewatering models, improved settler models, inclusion of soluble microbial products formation and extensions for physico-chemical processes. Furthermore, the BSM structure provides a framework to test models under development (e.g., Guo & Vanrolleghem, 2014).

10.3.5 Control strategy extensions

Advances in instrumentation and automation allow us to have access to highly accurate information regarding the urban wastewater system in real-time. Not only can this information be acquired from the WWTP, but on-line sensors can be installed in the sewer systems, and monitoring stations are being developed for monitoring river water quality (Copp *et al.* 2010). These developments are producing large quantities of data that can be used for fault-tolerant, uncertainty-aware and system-wide control design (Olsson *et al.* 2005; Olsson, 2012). Certainly, the benchmark platforms will be used in the future to test such new control strategies and the Task Group hopes that these new trends in process control, applied to the benchmark platforms, might contribute to our understanding on how such new control strategies can contribute to improving the operation of full-scale WWTPs.

10.3.6 Extended evaluation tools

The performance evaluation in the current benchmark platforms is based on three main types of evaluation criteria (effluent quality, operational cost issues and process risks). Effluent quality is considered through an effluent quality index (EQI), costs for energy, sludge and chemicals are considered through an operating cost index (OCI) and process risk is considered through a fuzzy logic calculation of microbiology-related operational problems to create a risk index. The wish list for future expansions of the evaluation criteria contains energy consumption models, validation and extension of the risk calculation concerning microbiology-related TSS separation problems, inclusion of capital and maintenance costs, robust uncertainty based evaluation of control strategies (e.g., the robustness index proposed by Vanrolleghem &

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Gillot, 2002), as well as a set of criteria to capture greenhouse gas emissions and the carbon footprint of the WWTP (e.g., the proposal discussed in Corominas *et al*. 2012).

10.4 THE 'BENCHMARKING SPIRIT'

Two central elements in the benchmarking community philosophy have been the development of validated models and the free distribution of the verified models. All of this has been possible because of a tremendous amount of voluntary work. With over 400 scientific papers published over the years, the results of that voluntary effort are very impressive.

It is the Task Group's sincere hope that this report will inspire research groups to carry on the development effort allowing it to flourish and continue to be a state-of-the-art tool for research, model development and practical application of control within the field of wastewater treatment. This STR is the culmination of this stage in the BSM development, but with so many possibilities for expansion and improvement, hopefully a new wave of motivated scientists will foster new like-minded life-long friendships and continue the development well into the future.

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